

BFS Traversal

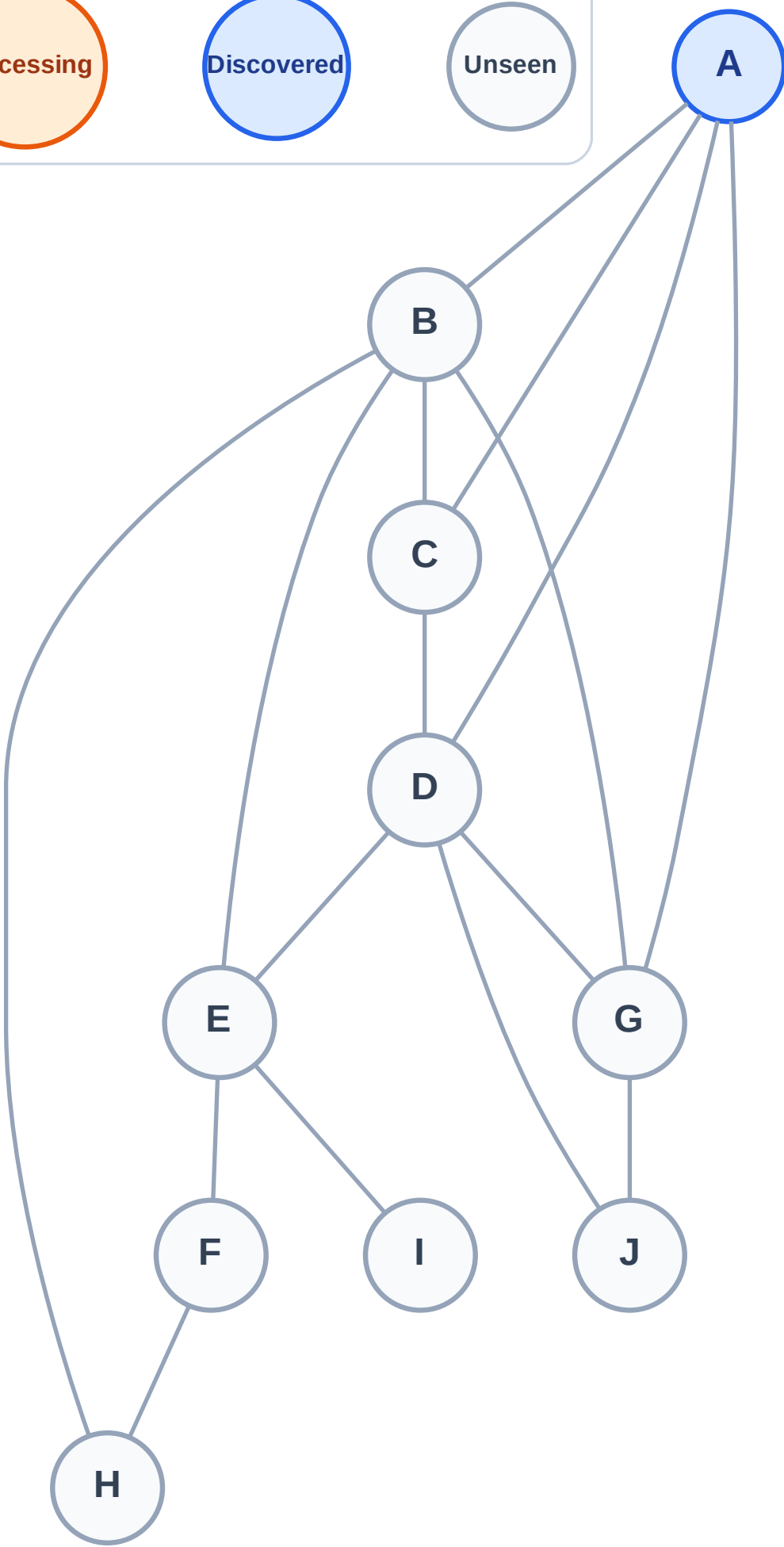
Step 1: Start at A

Legend



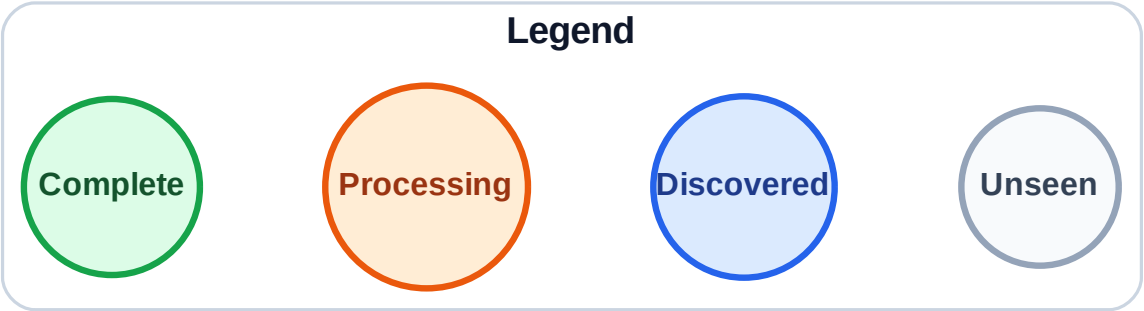
Queue: A

Visited: (none)

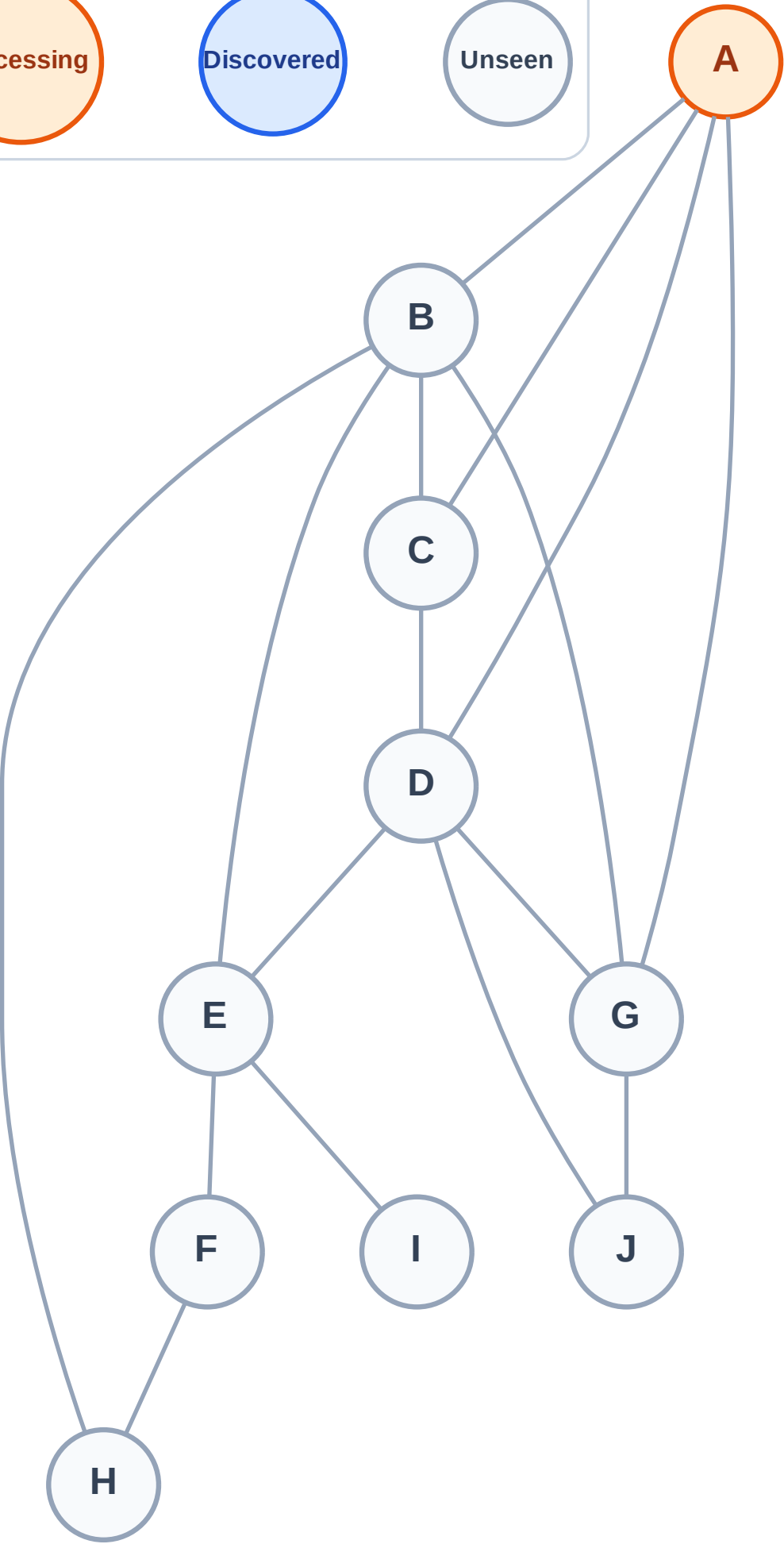


BFS Traversal

Step 2: Process node A



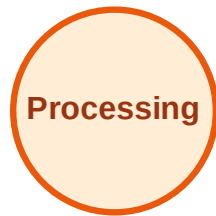
Queue: (empty)
Visited: (none)



BFS Traversal

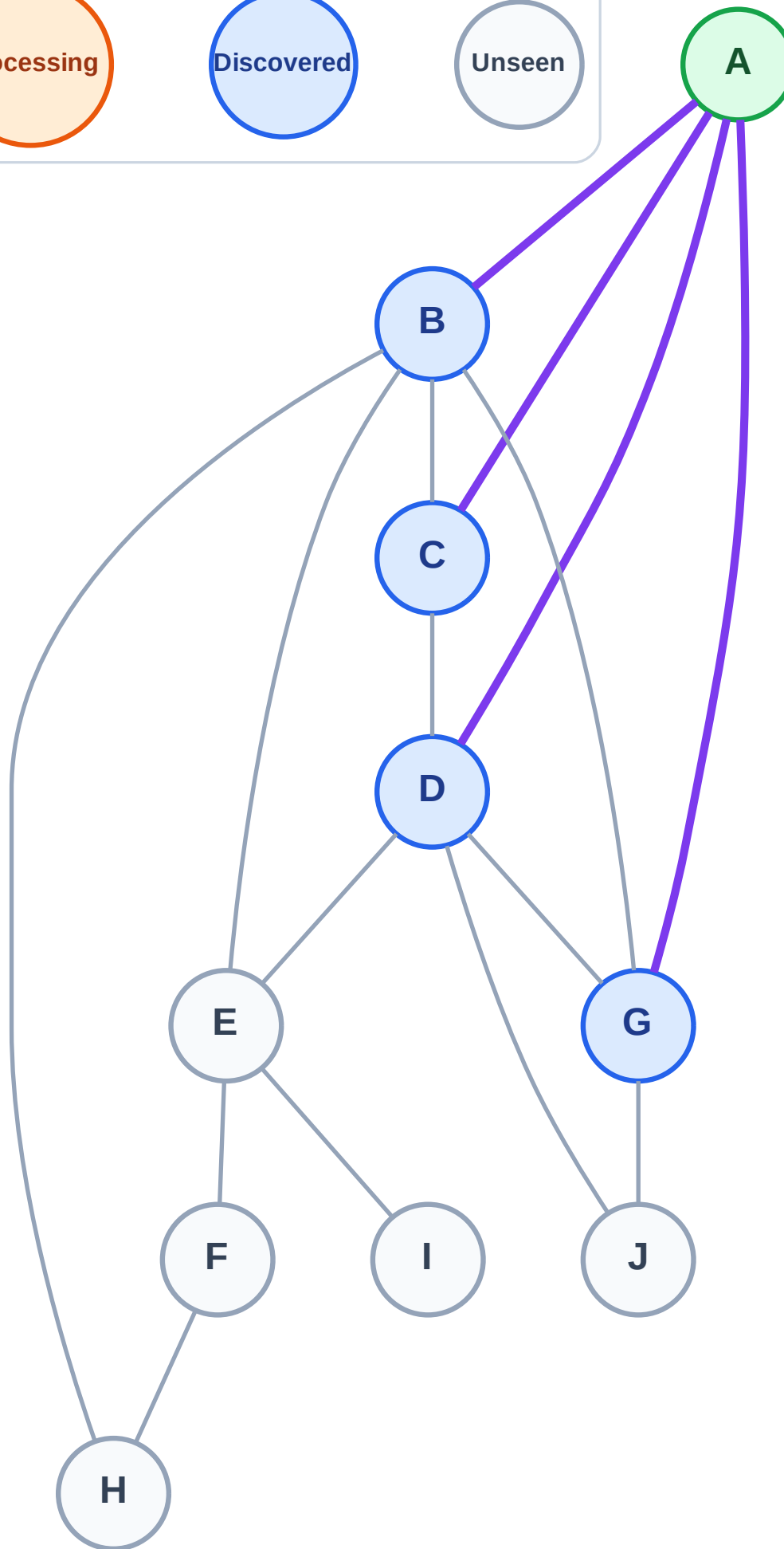
Step 3: Discover B, C, D, G from A

Legend



Queue: B → C → D → G

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

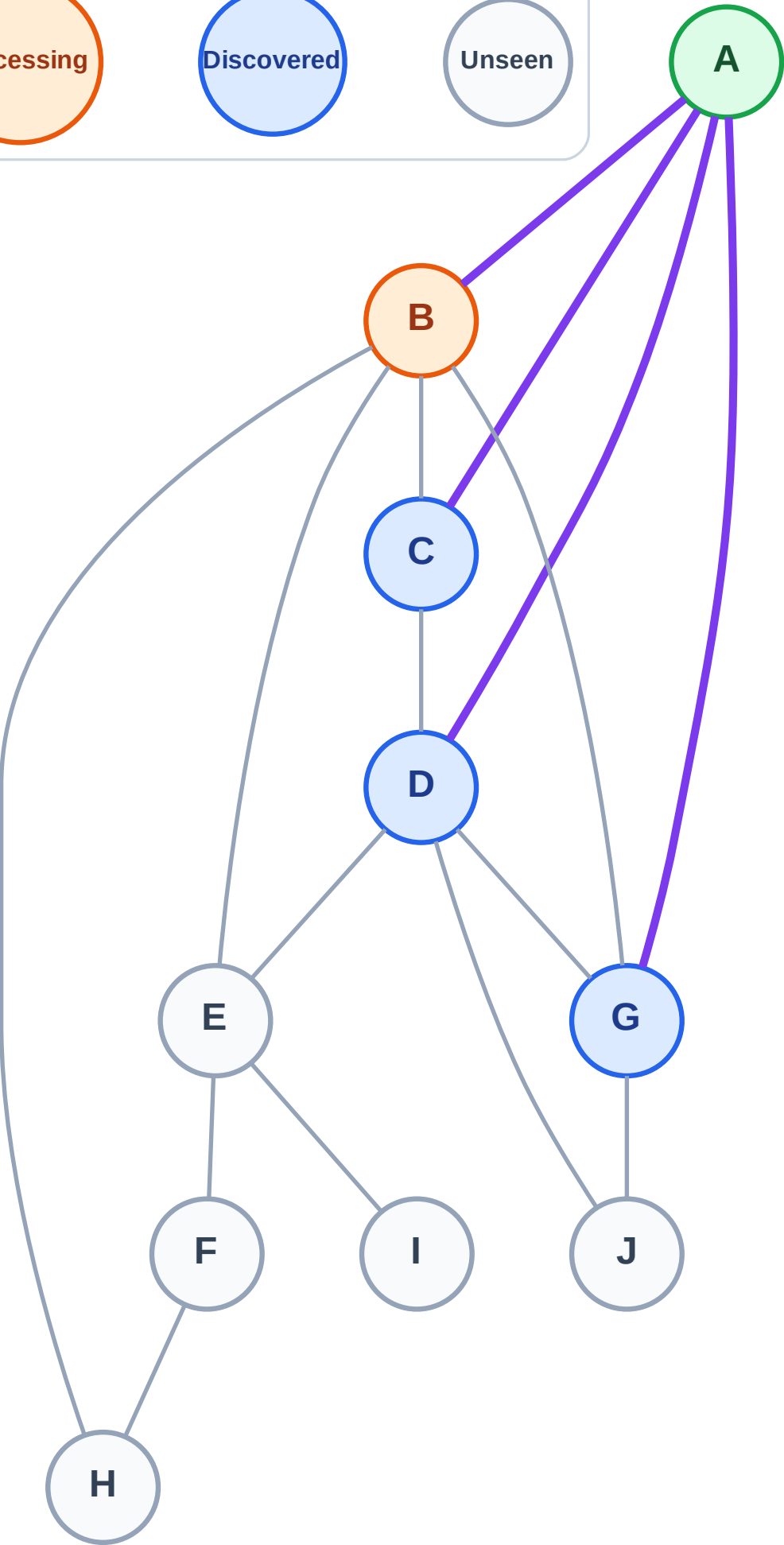
Processing

Discovered

Unseen

Queue: C → D → G

Visited: A



BFS Traversal

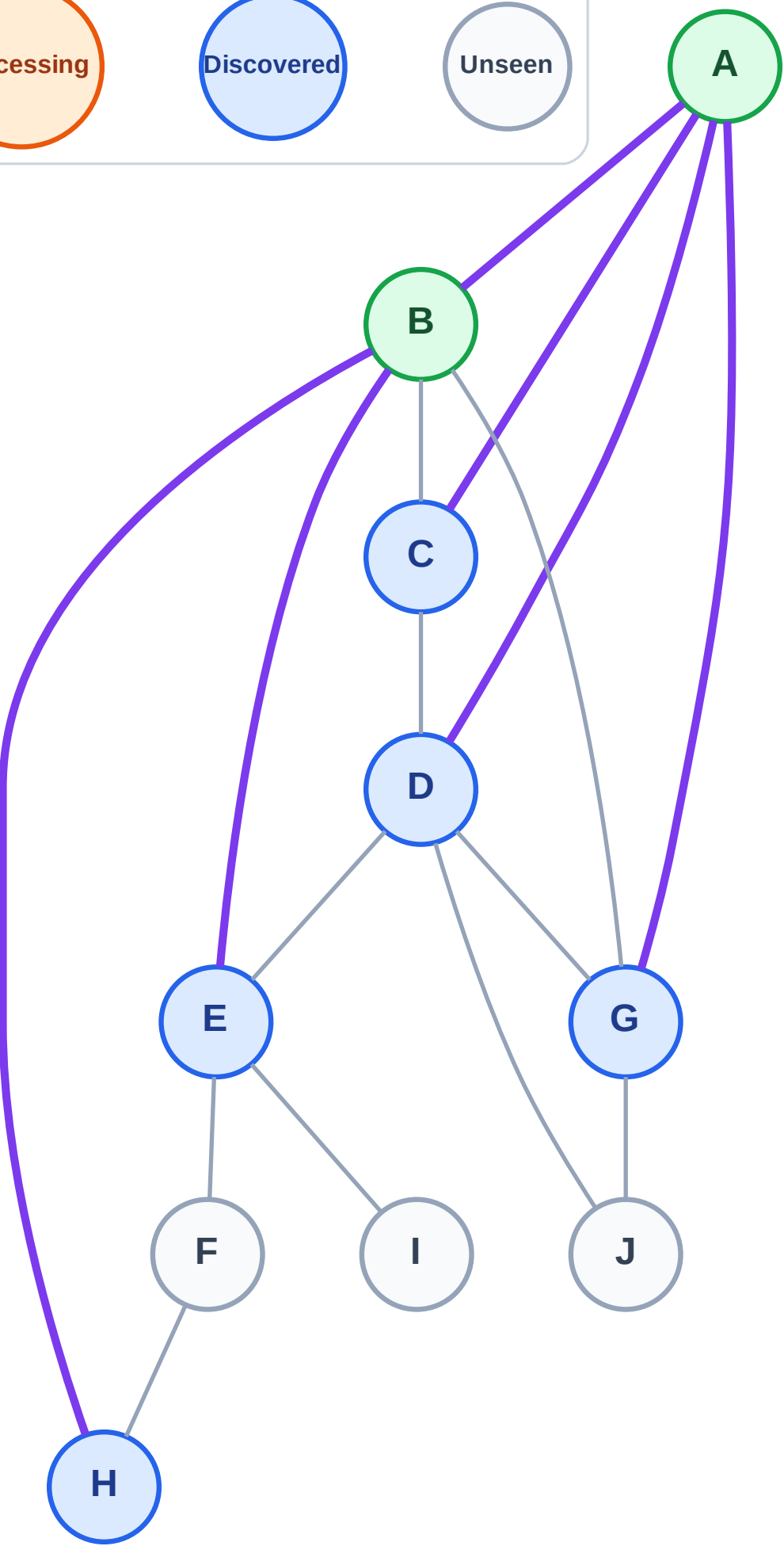
Step 5: Discover E, H from B

Legend



Queue: C → D → G → E → H

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

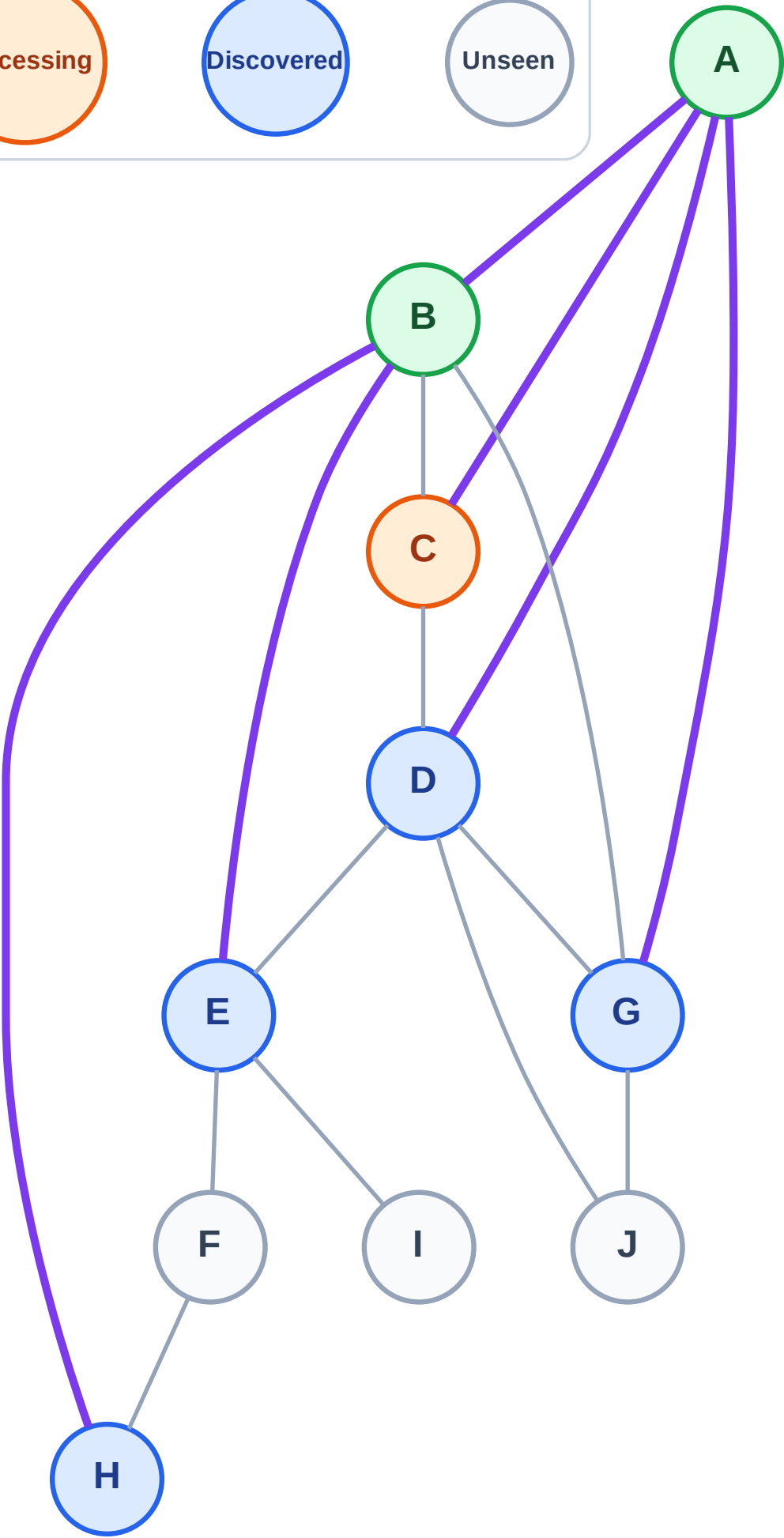
Processing

Discovered

Unseen

Queue: D → G → E → H

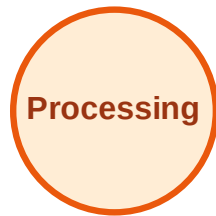
Visited: A, B



BFS Traversal

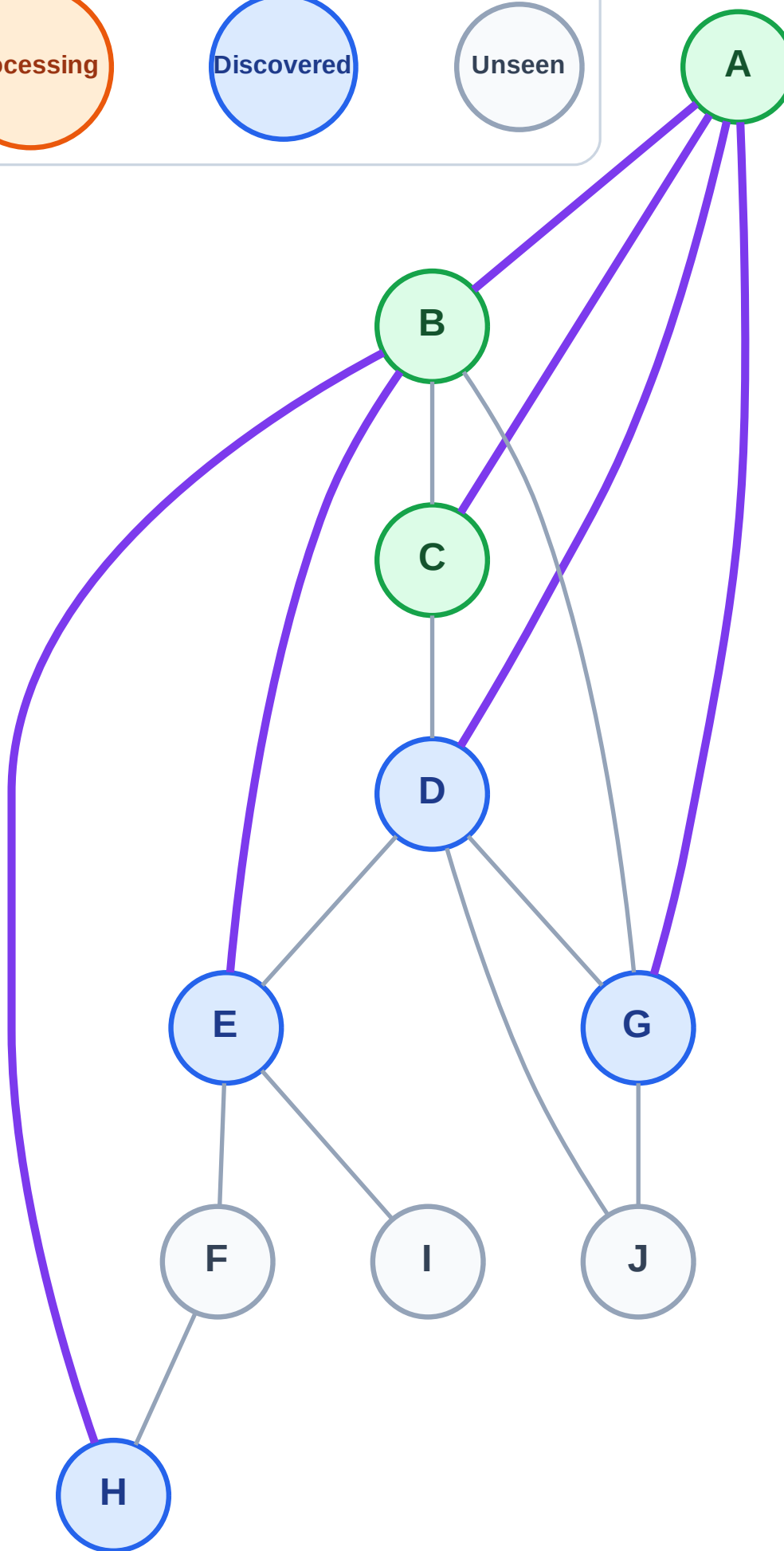
Step 7: No new neighbors from C

Legend



Queue: D → G → E → H

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

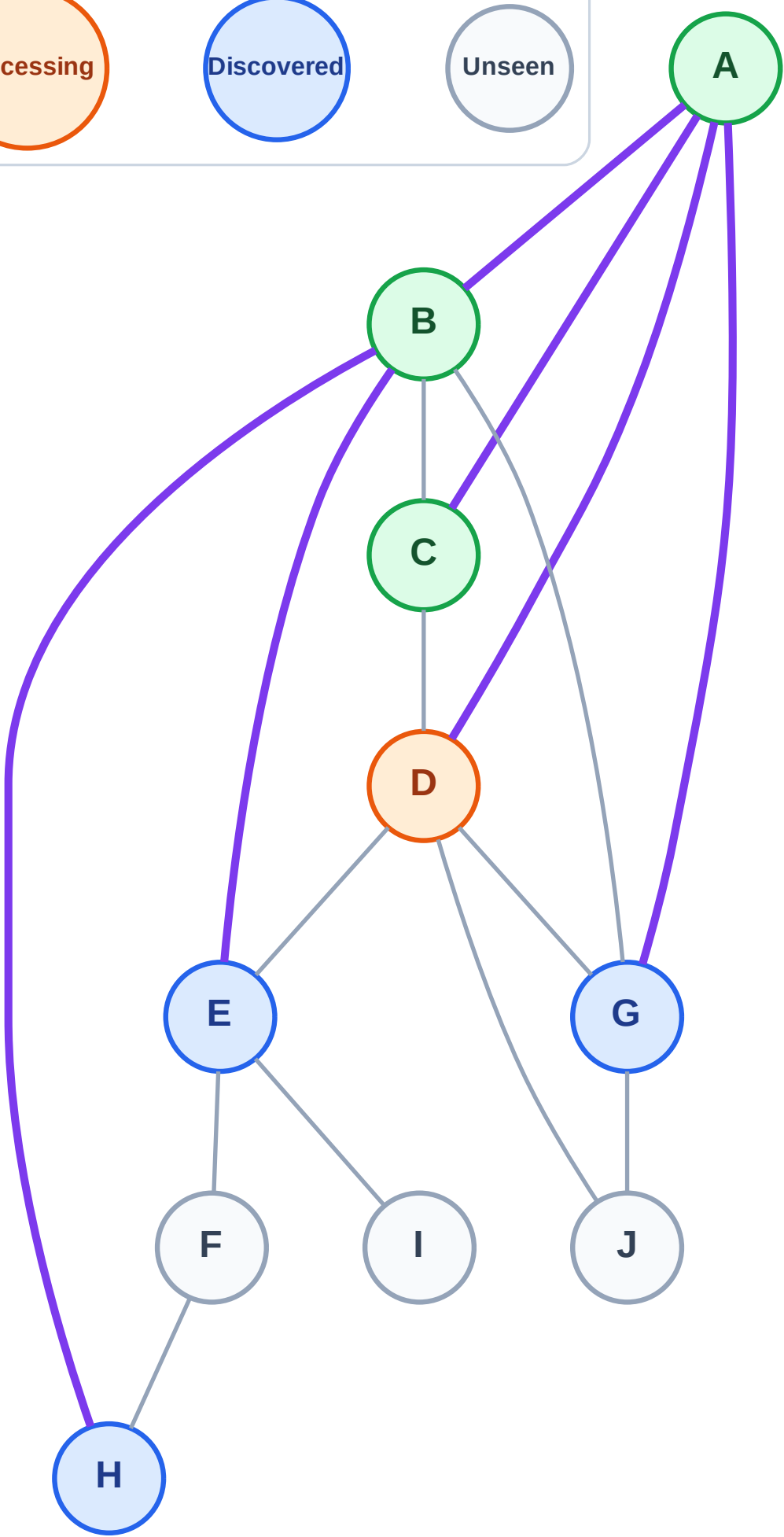
Processing

Discovered

Unseen

Queue: G → E → H

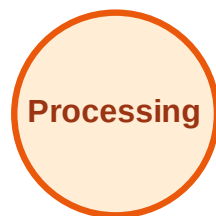
Visited: A, B, C



BFS Traversal

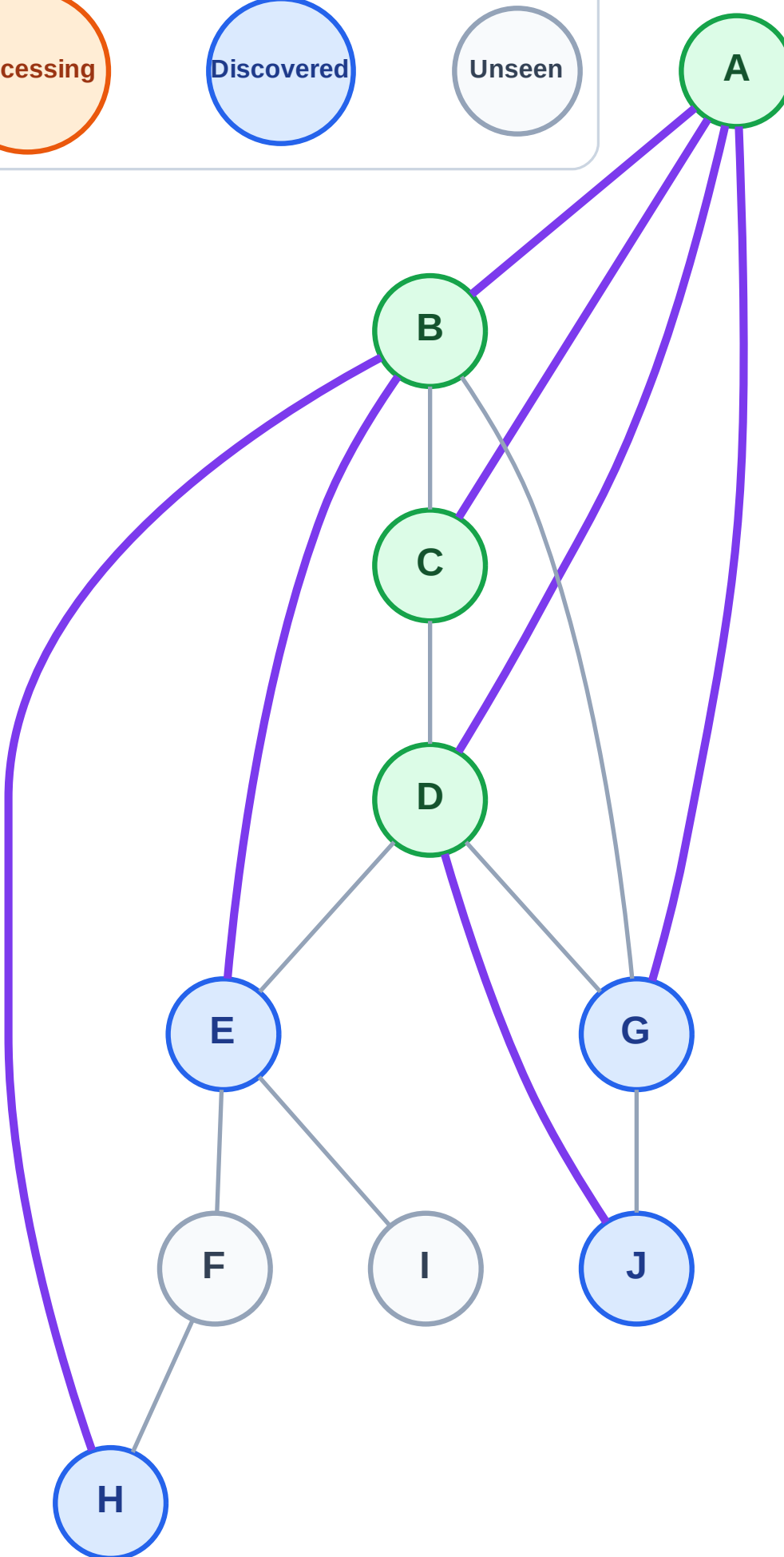
Step 9: Discover J from D

Legend



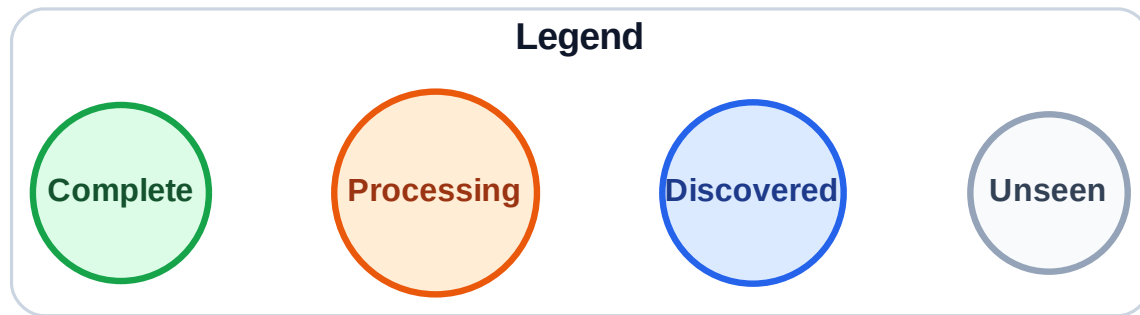
Queue: G → E → H → J

Visited: A, B, C, D

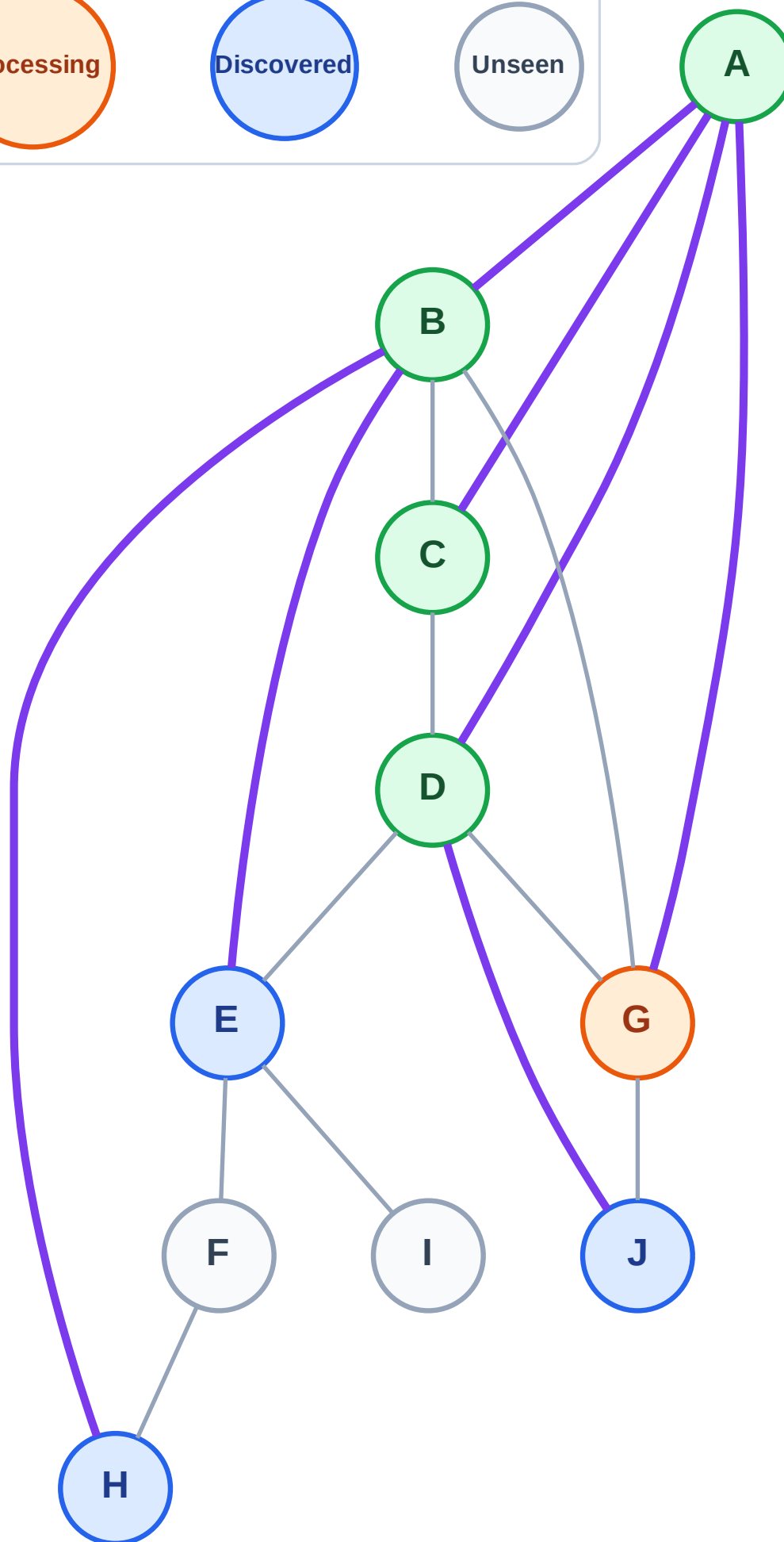


BFS Traversal

Step 10: Process node G

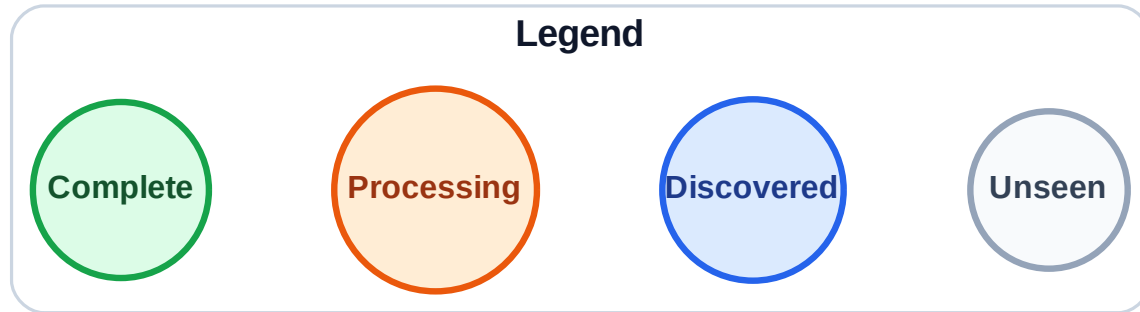


Queue: E → H → J
Visited: A, B, C, D

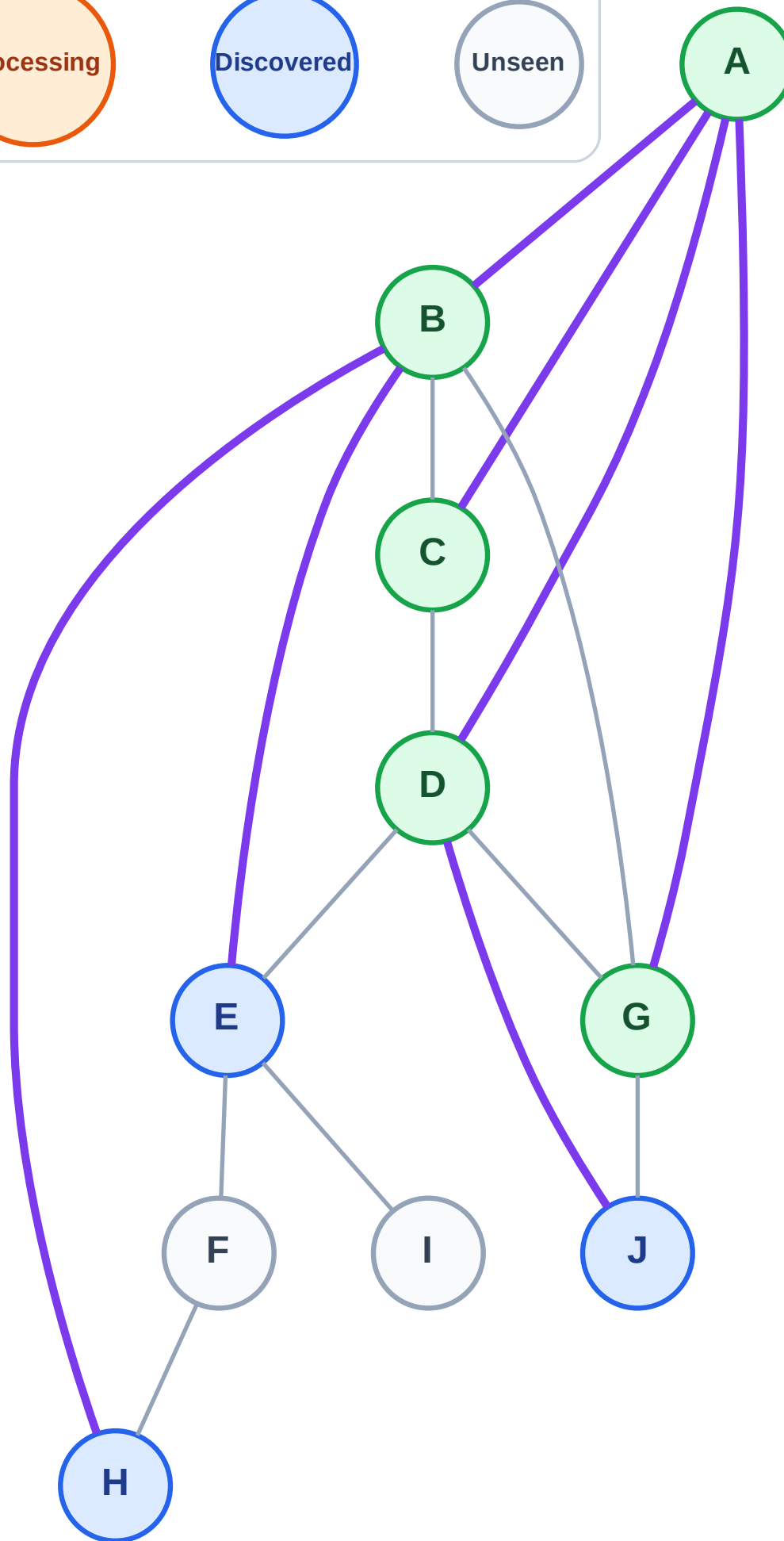


BFS Traversal

Step 11: No new neighbors from G

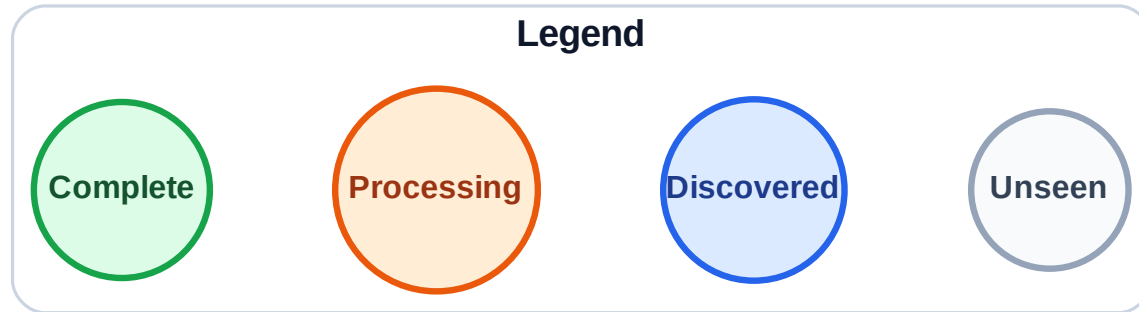


Queue: E → H → J
Visited: A, B, C, D, G

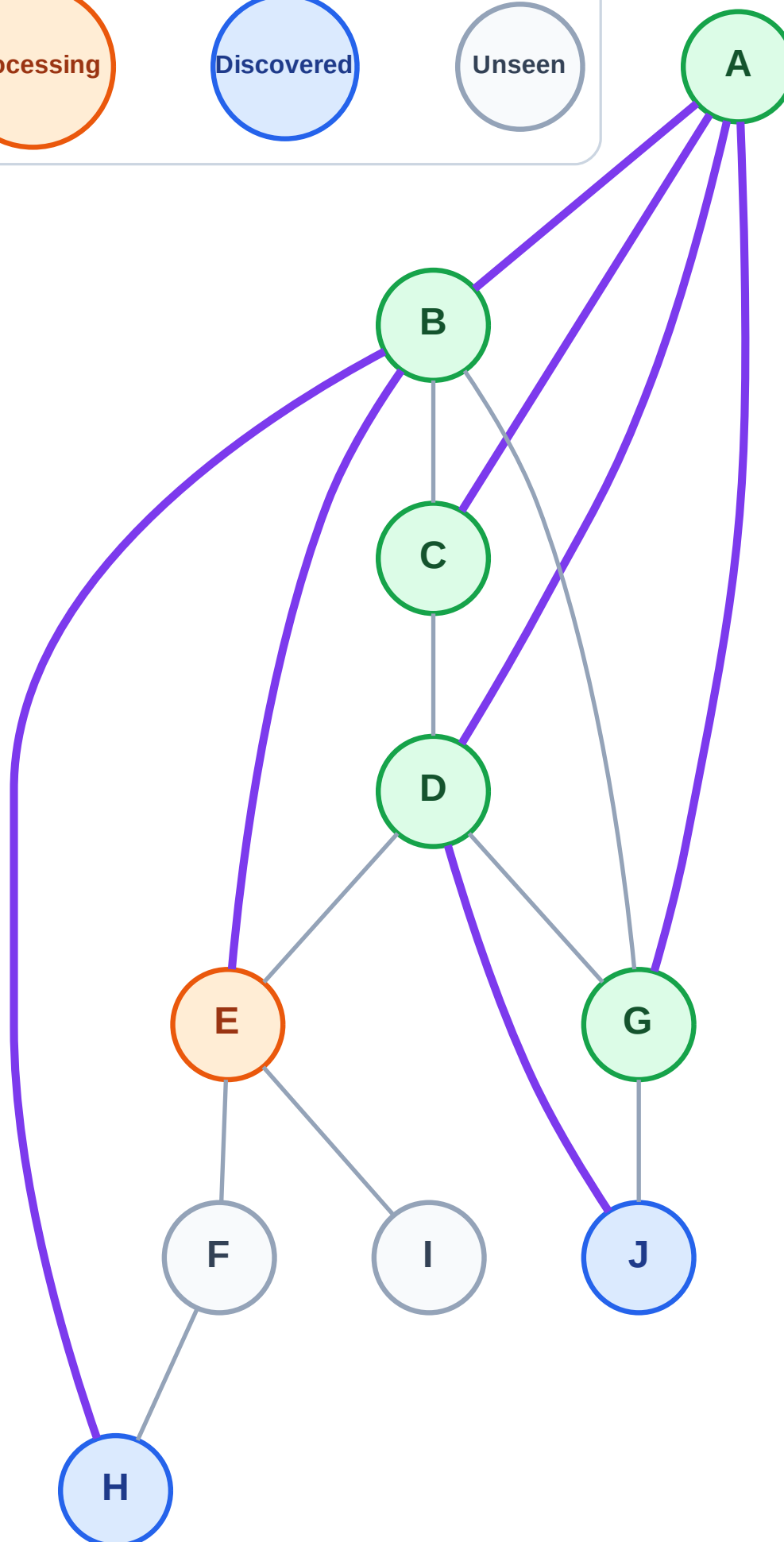


BFS Traversal

Step 12: Process node E



Queue: H → J
Visited: A, B, C, D, G



BFS Traversal

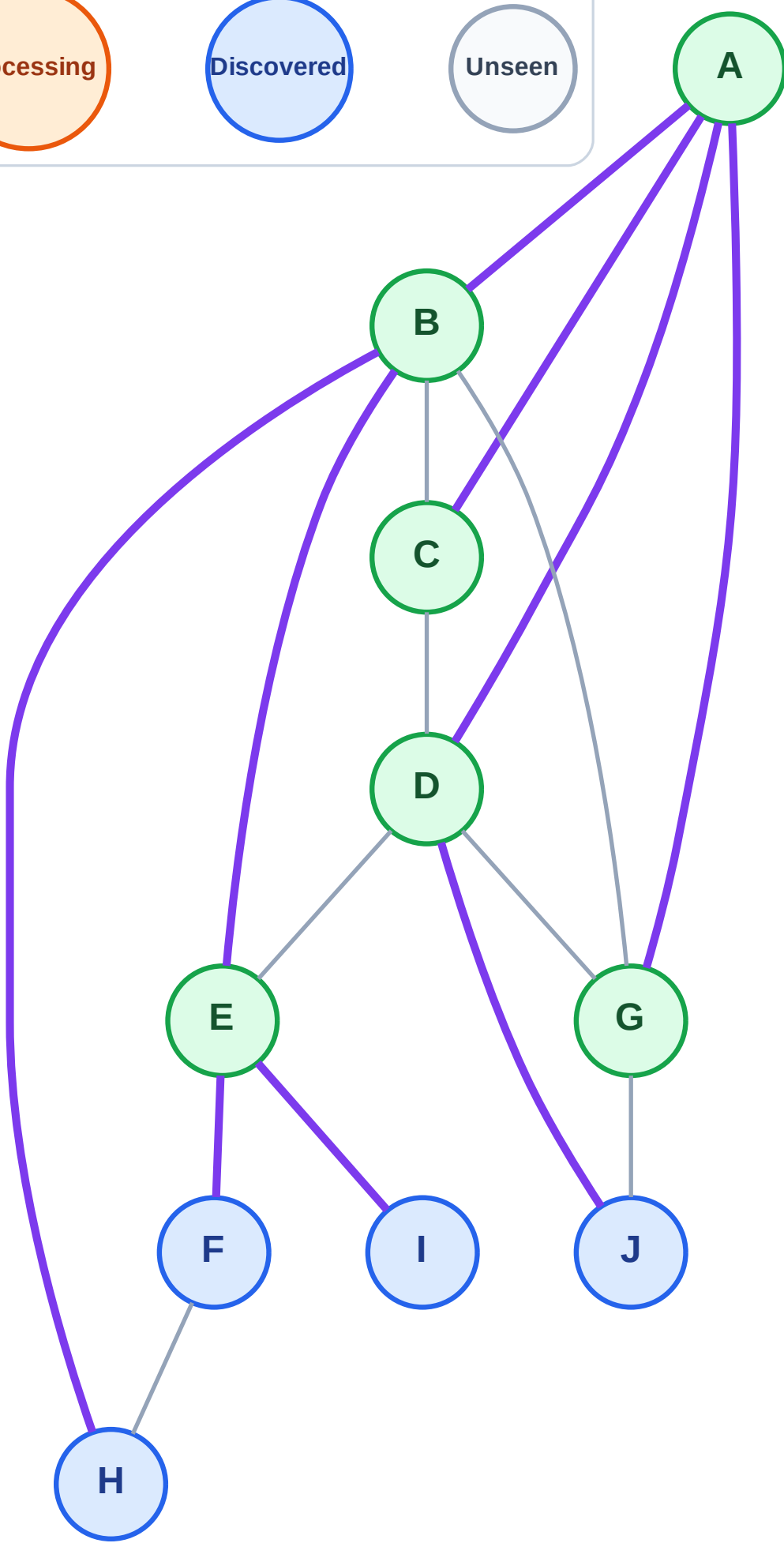
Step 13: Discover F, I from E

Legend



Queue: H → J → F → I

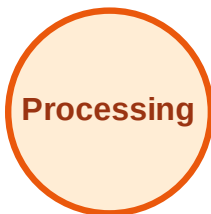
Visited: A, B, C, D, E, G



BFS Traversal

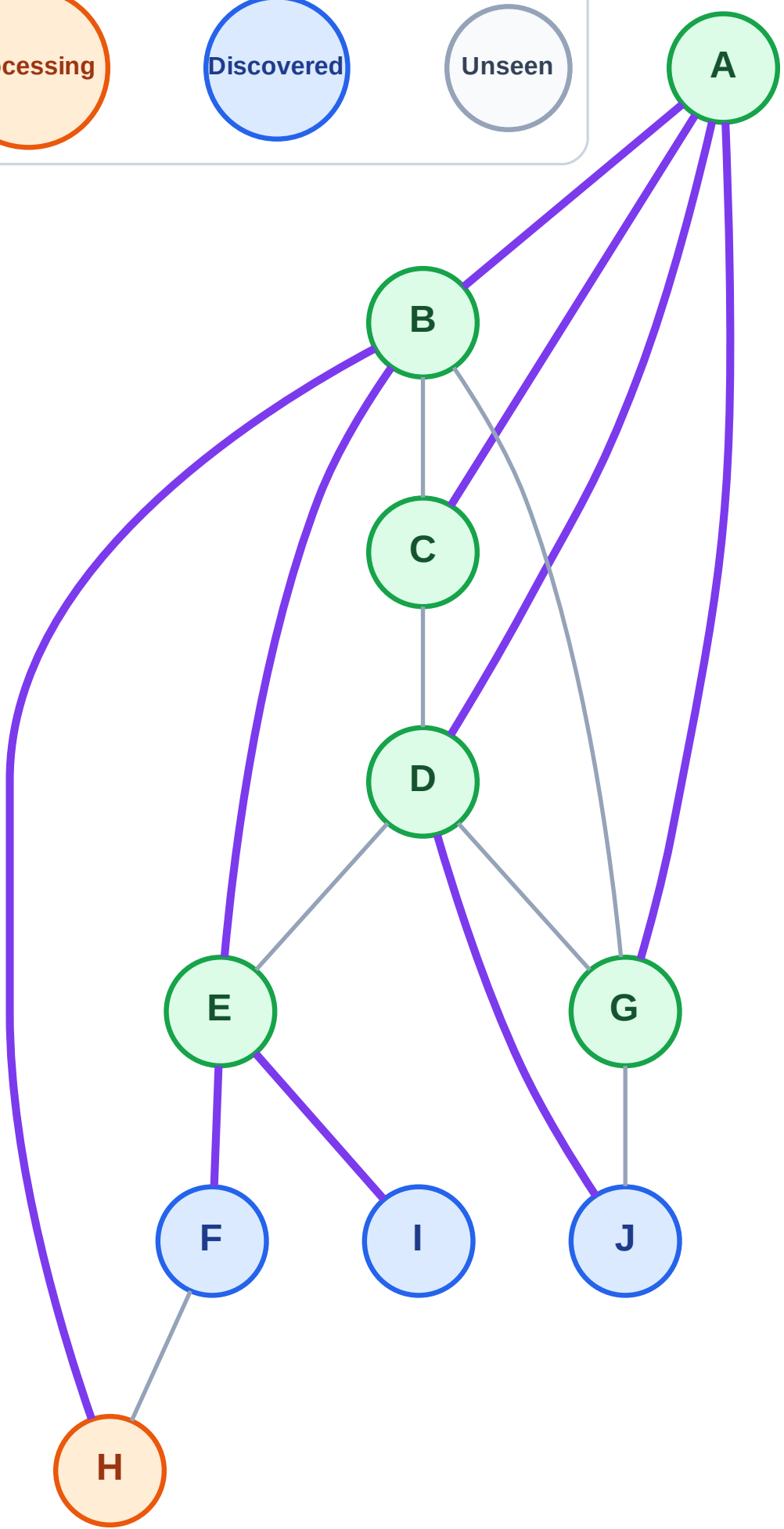
Step 14: Process node H

Legend



Queue: J → F → I

Visited: A, B, C, D, E, G



BFS Traversal

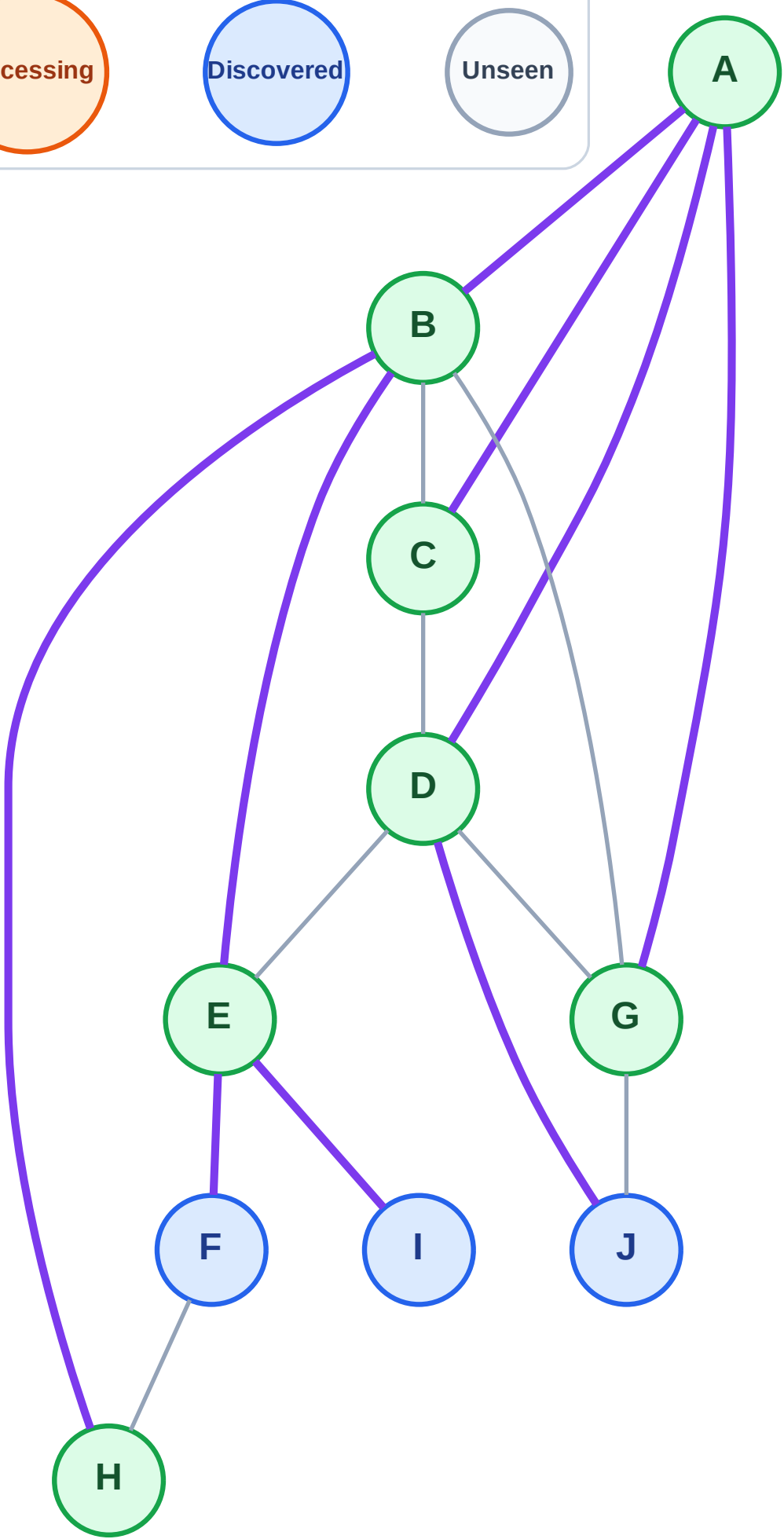
Step 15: No new neighbors from H

Legend



Queue: J → F → I

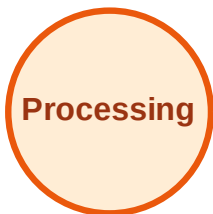
Visited: A, B, C, D, E, G, H



BFS Traversal

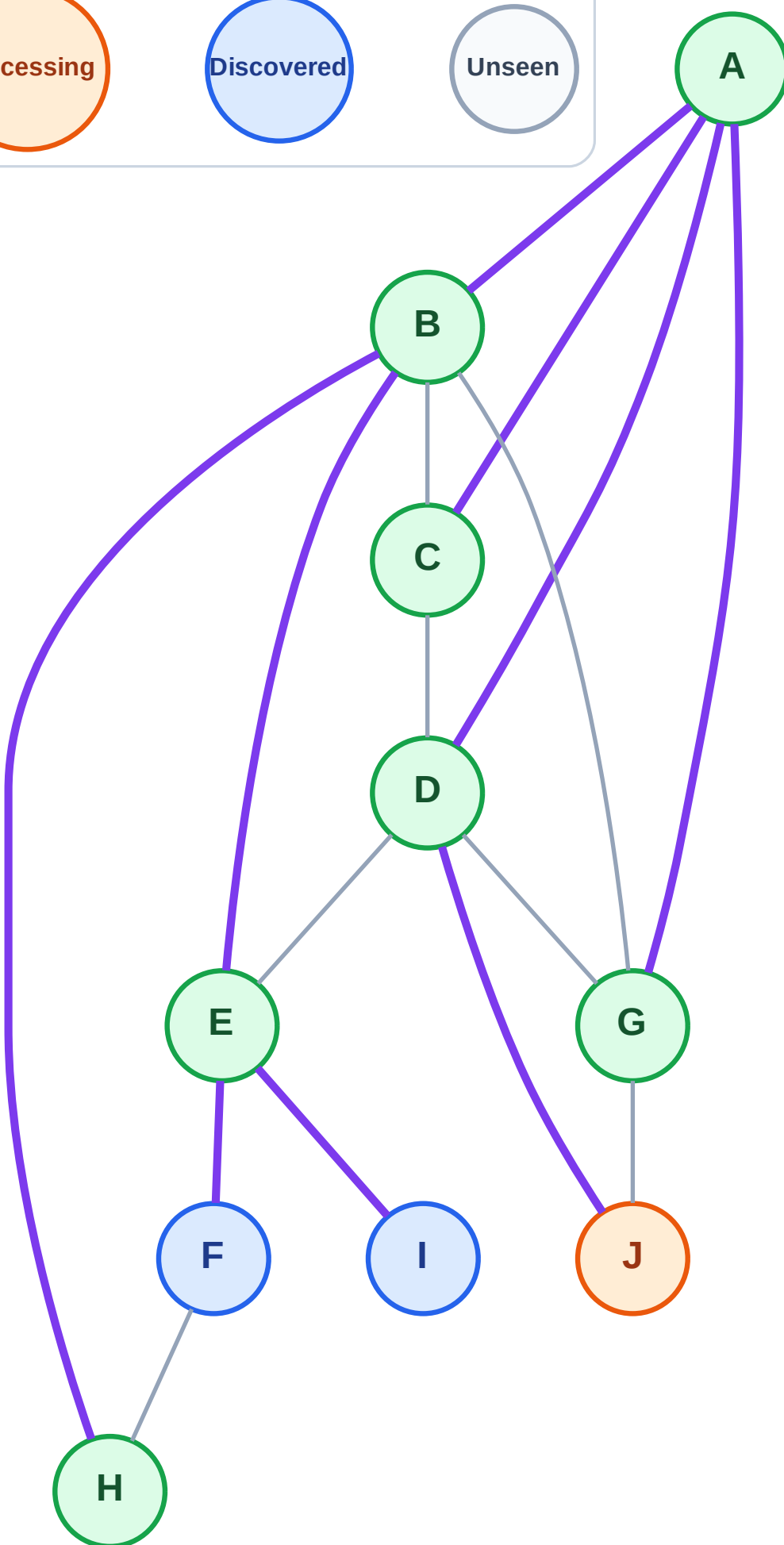
Step 16: Process node J

Legend



Queue: F → I

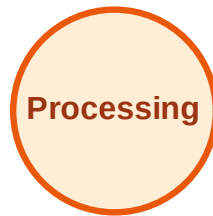
Visited: A, B, C, D, E, G, H



BFS Traversal

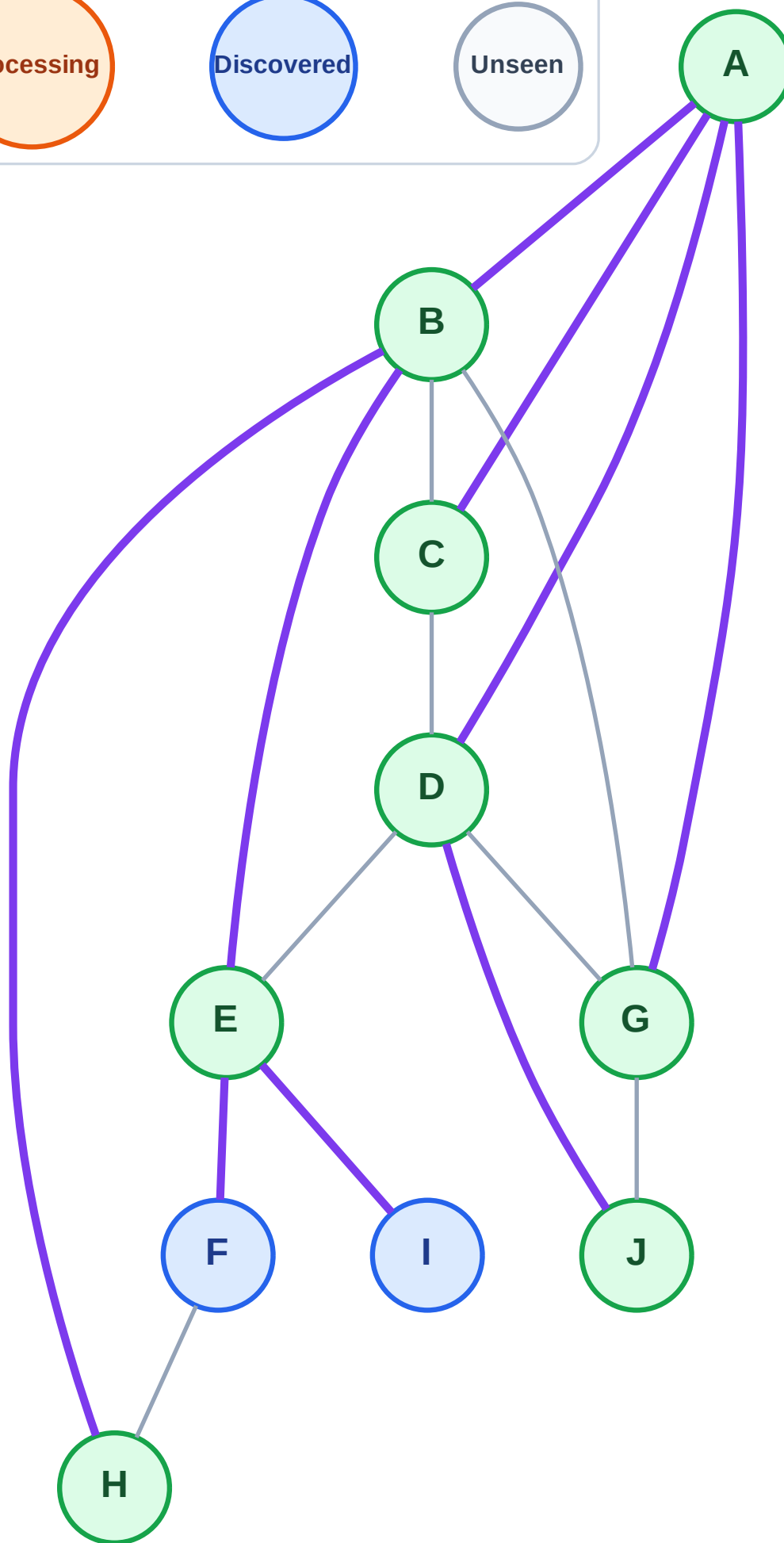
Step 17: No new neighbors from J

Legend



Queue: F → I

Visited: A, B, C, D, E, G, H, J



BFS Traversal

Step 18: Process node F

Legend

Complete

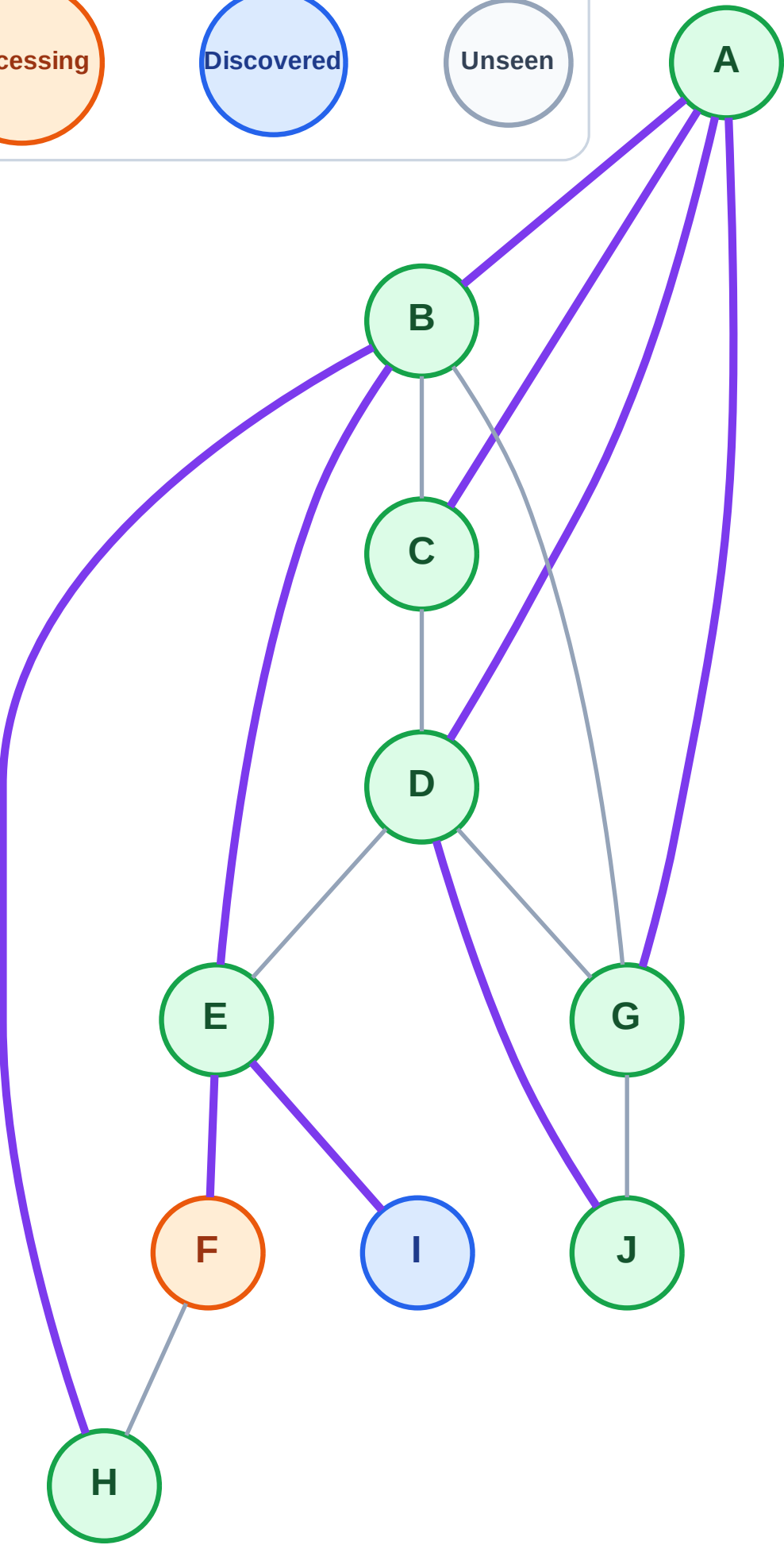
Processing

Discovered

Unseen

Queue: I

Visited: A, B, C, D, E, G, H, J



BFS Traversal

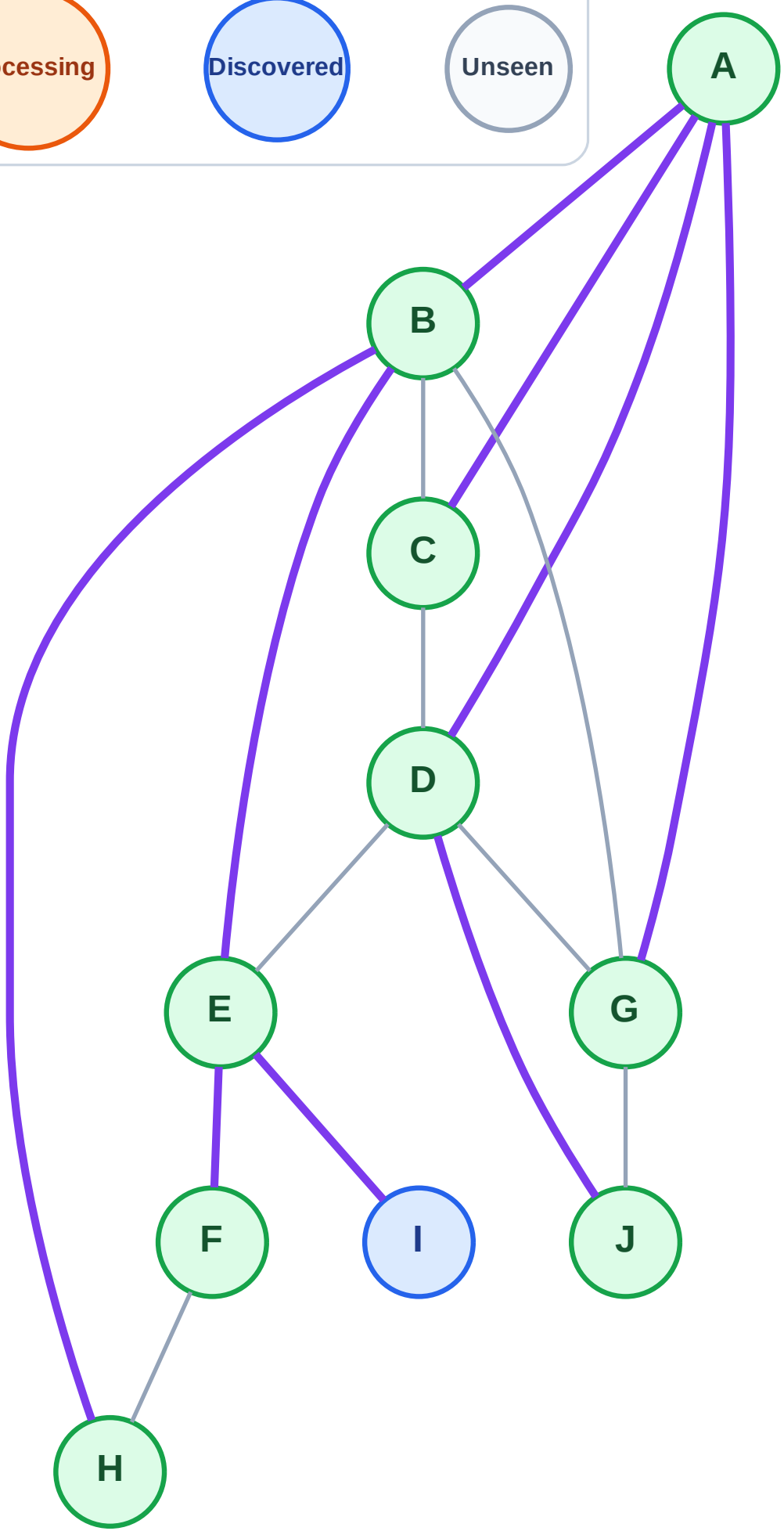
Step 19: No new neighbors from F

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

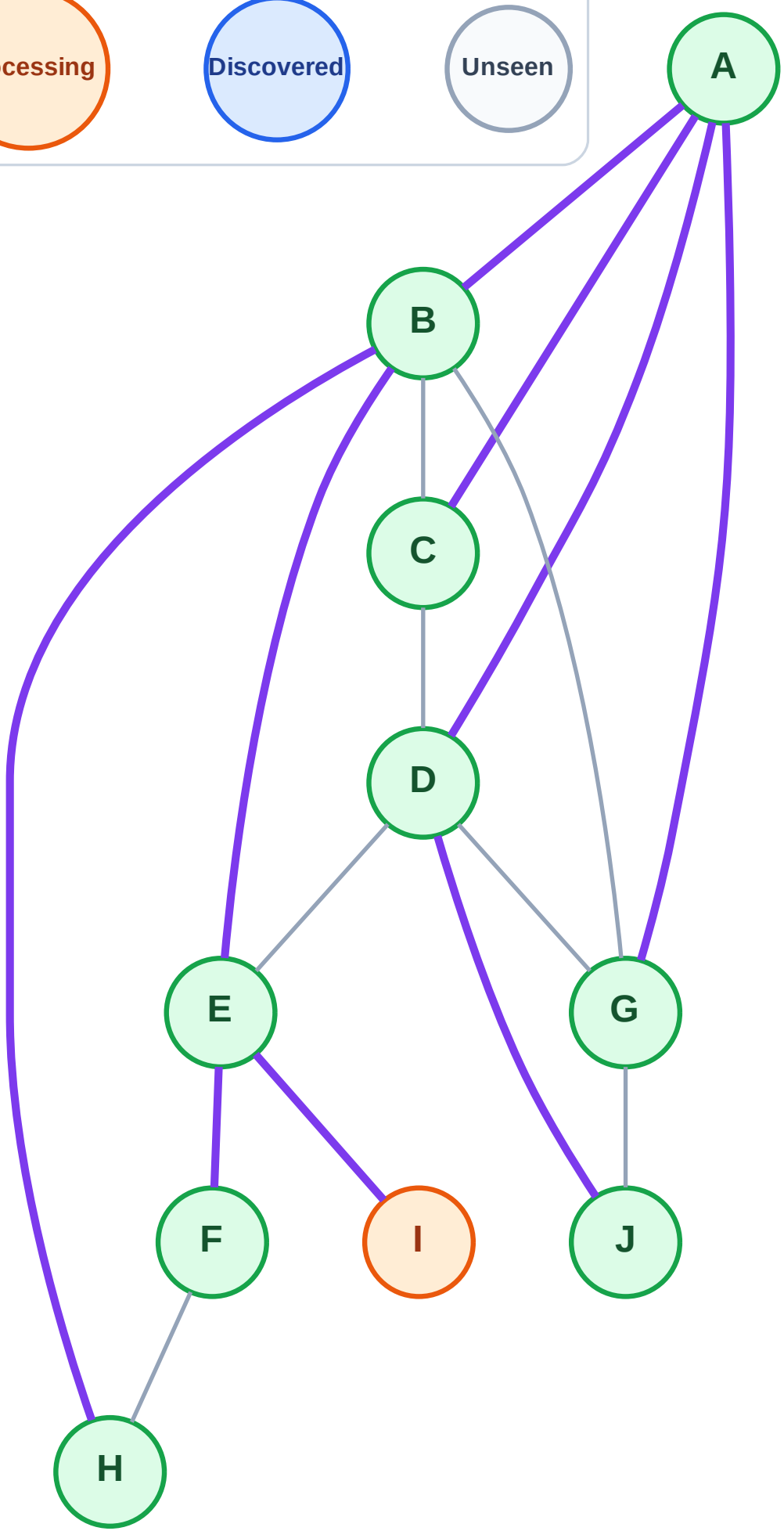
Step 20: Process node I

Legend



Queue: (empty)

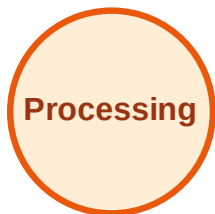
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

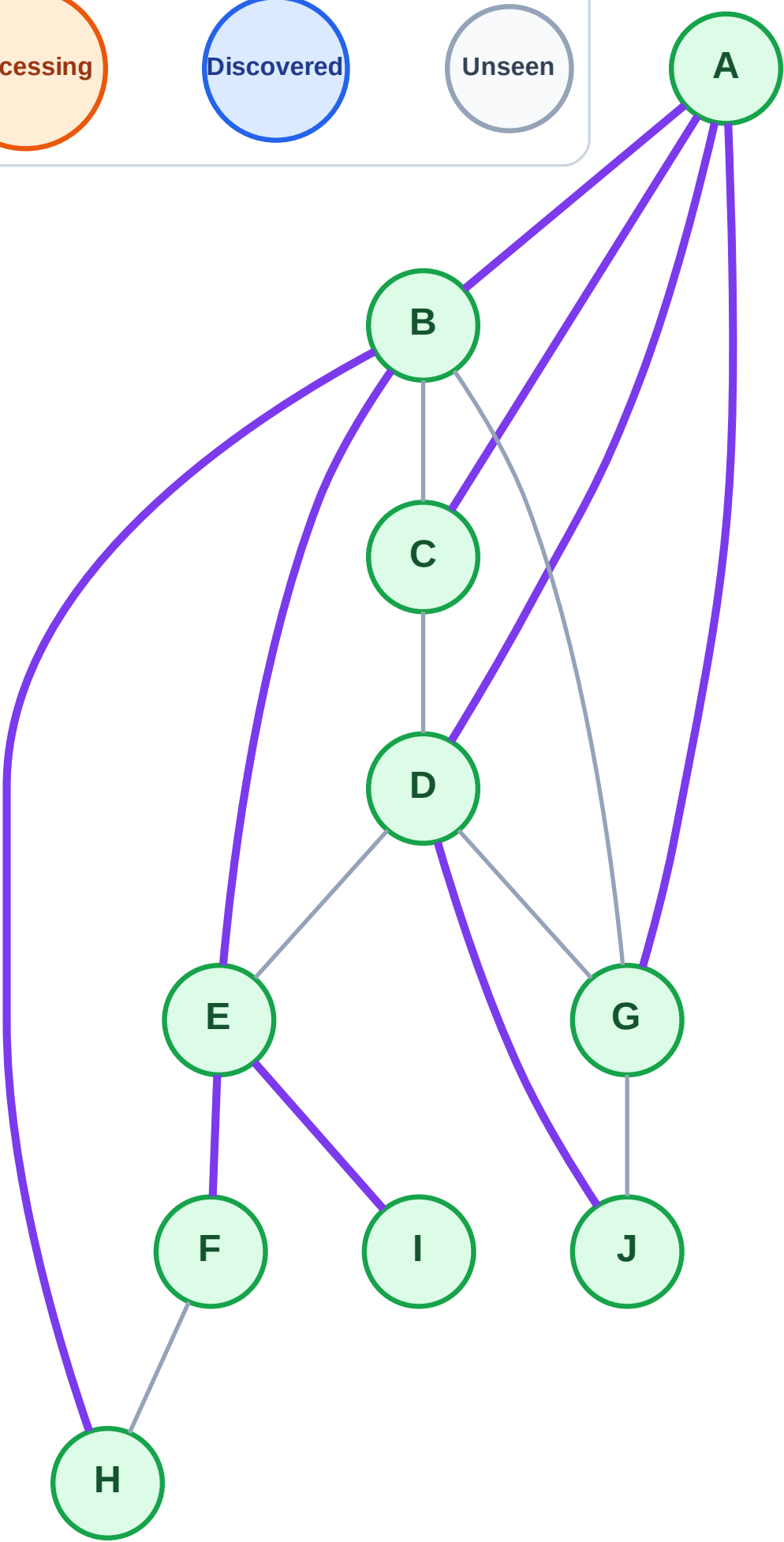
Step 21: No new neighbors from I

Legend



Queue: (empty)

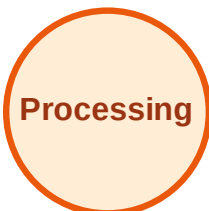
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

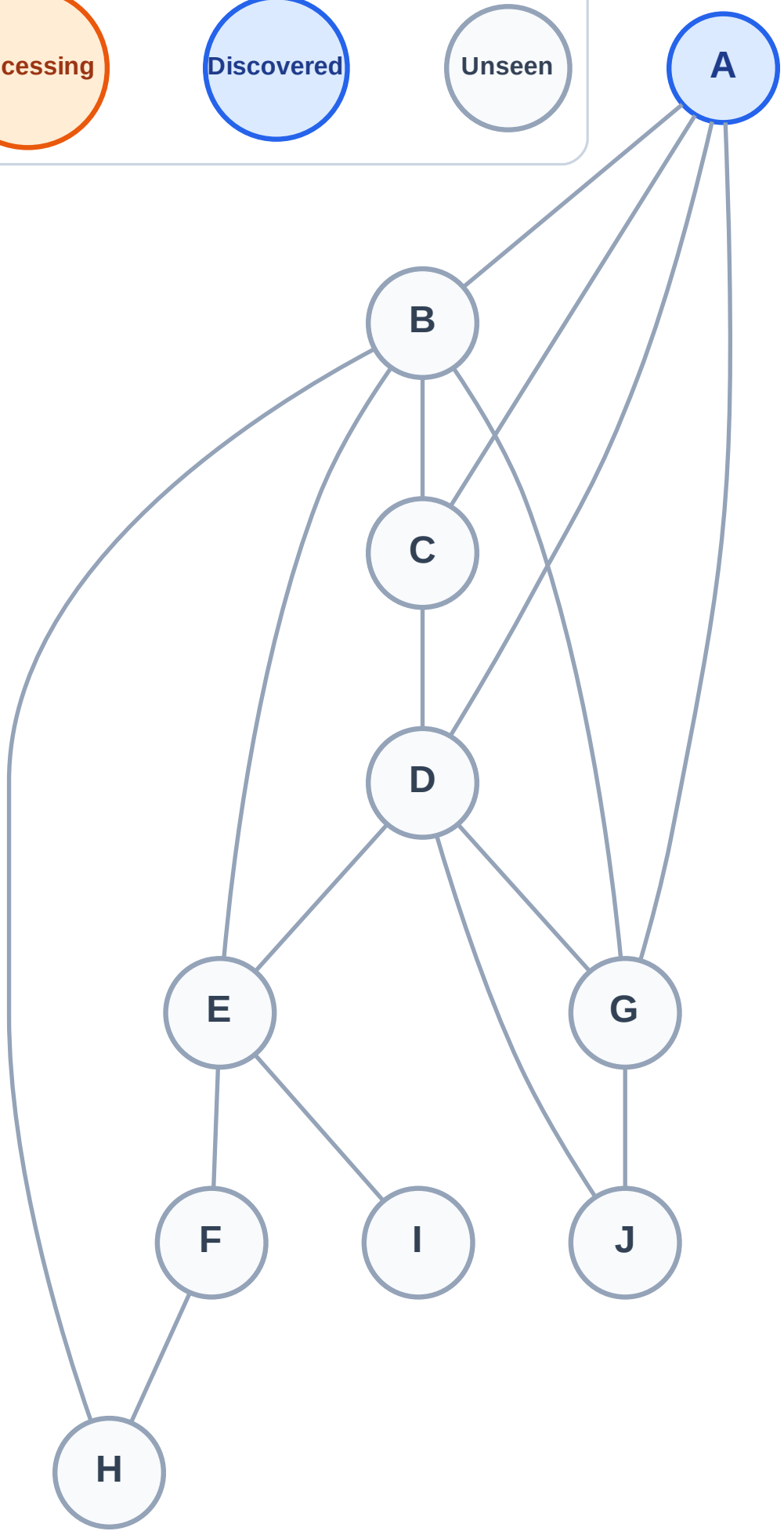
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

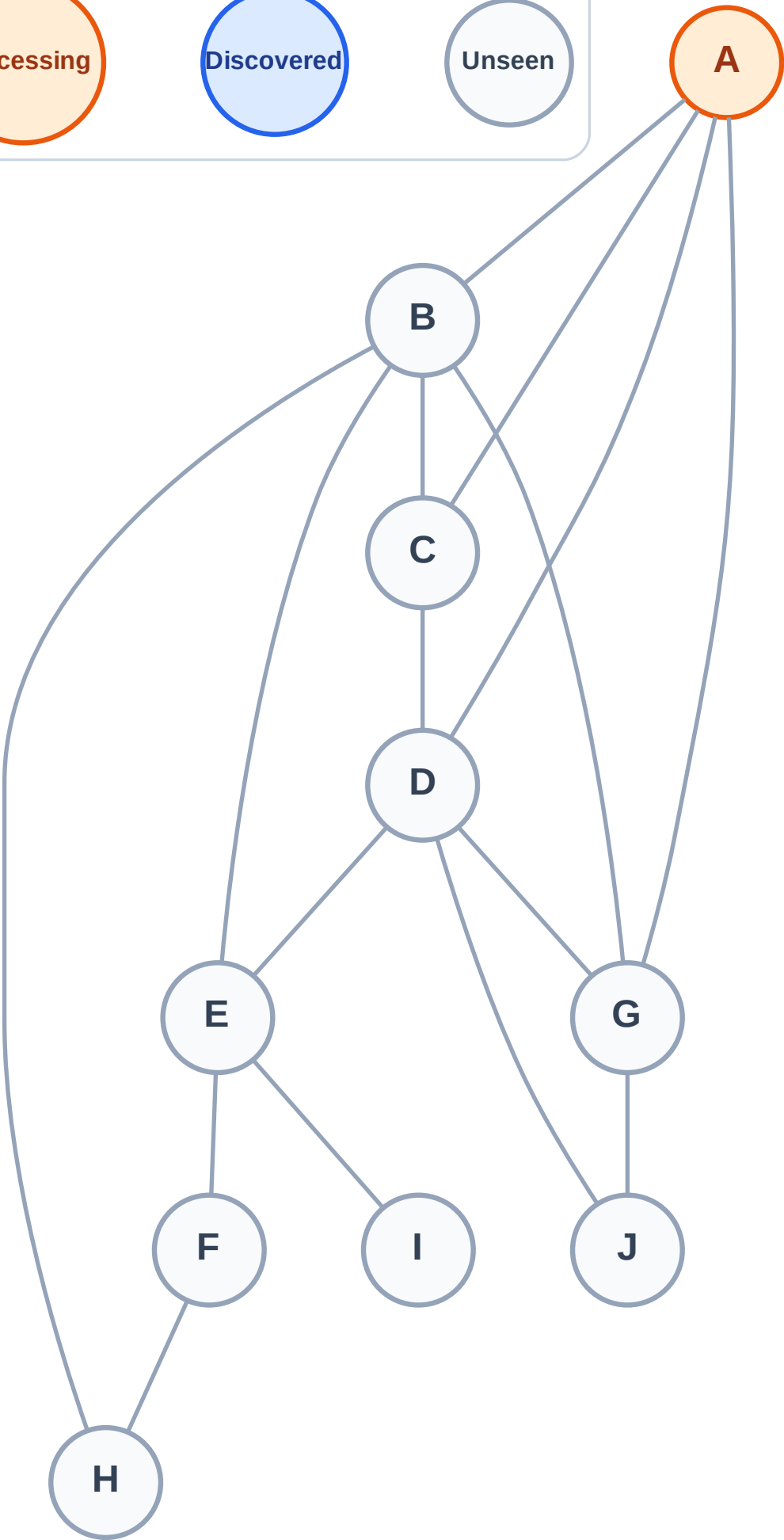
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover G, D, C, B from A

Legend

Complete

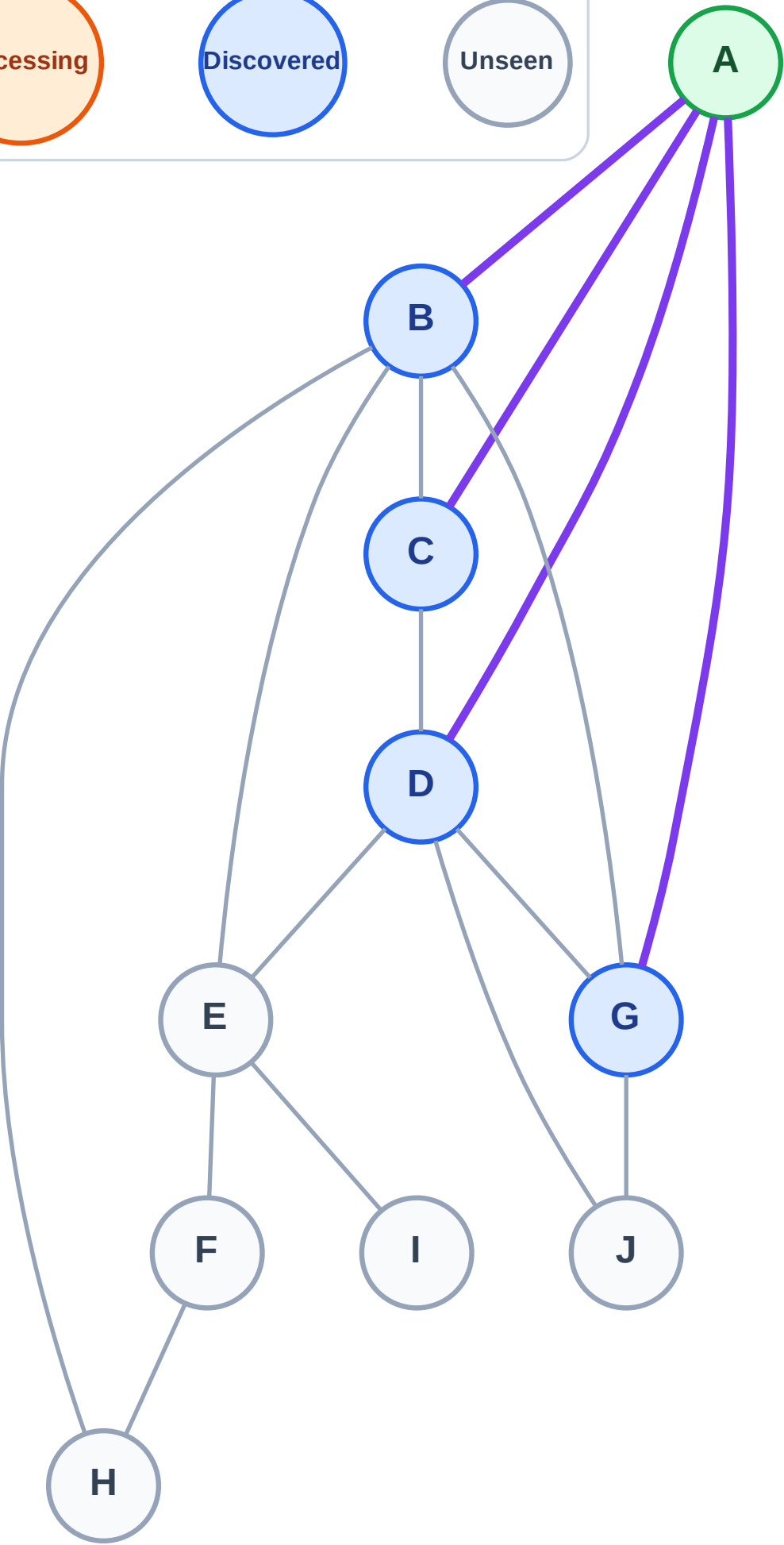
Processing

Discovered

Unseen

Stack (top at right): G | D | C | B

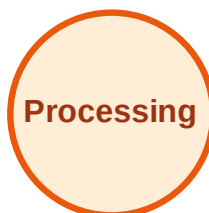
Visited: A



DFS Traversal

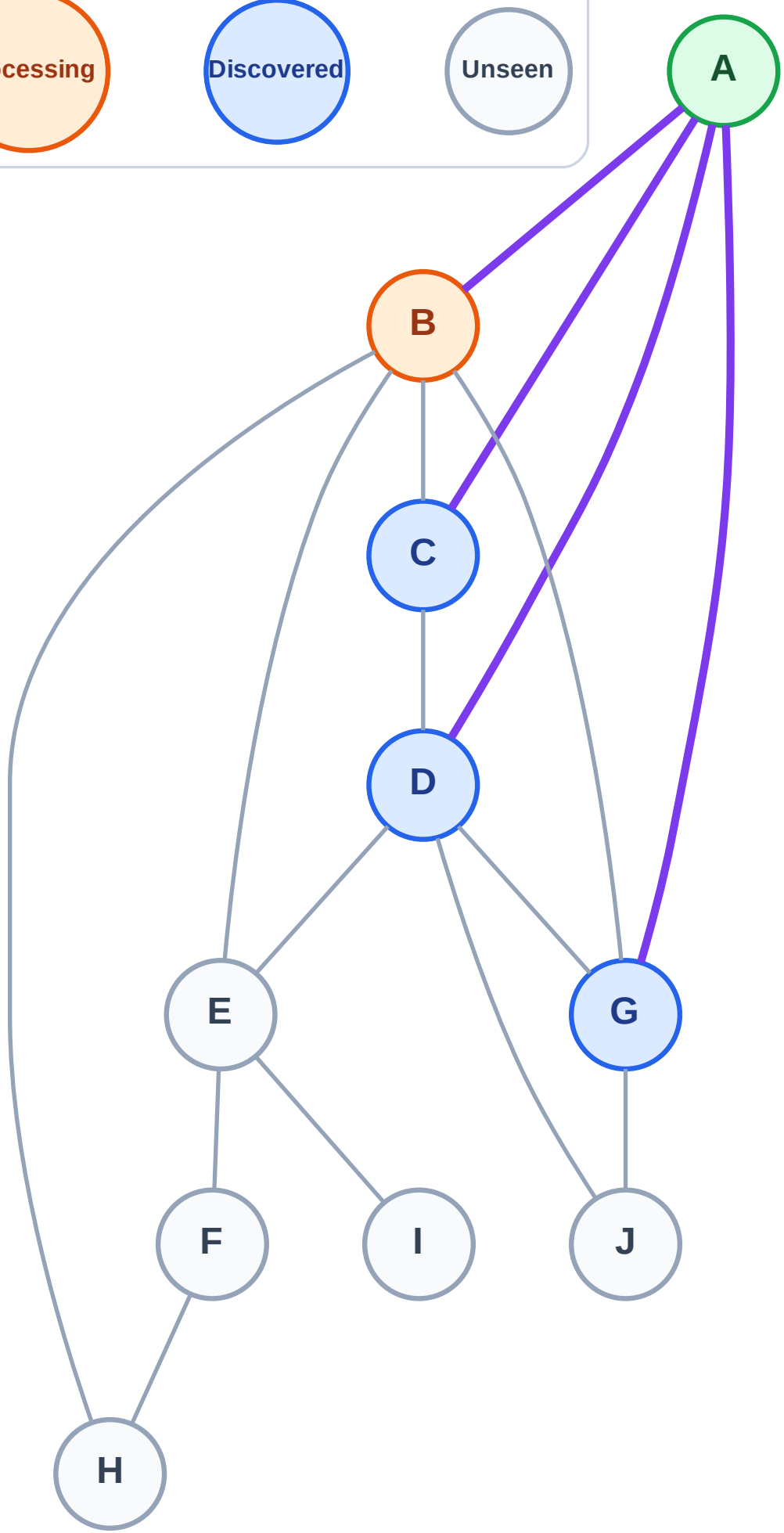
Step 4: Process node B

Legend



Stack (top at right): G | D | C

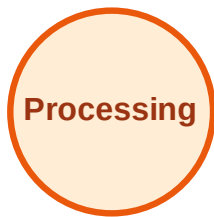
Visited: A



DFS Traversal

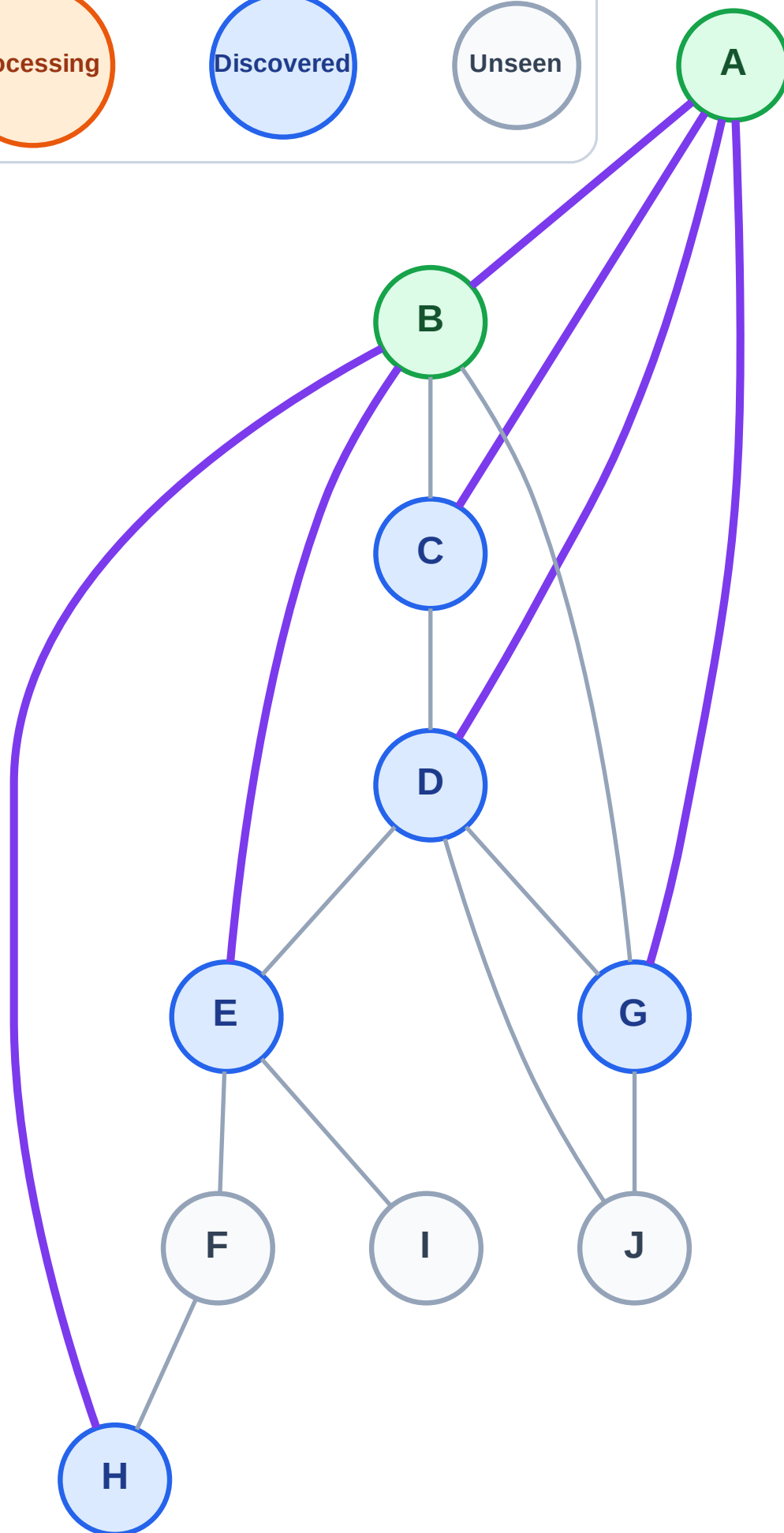
Step 5: Discover H, E from B

Legend



Stack (top at right): G | D | C | H | E

Visited: A, B



DFS Traversal

Step 6: Process node E

Legend

Complete

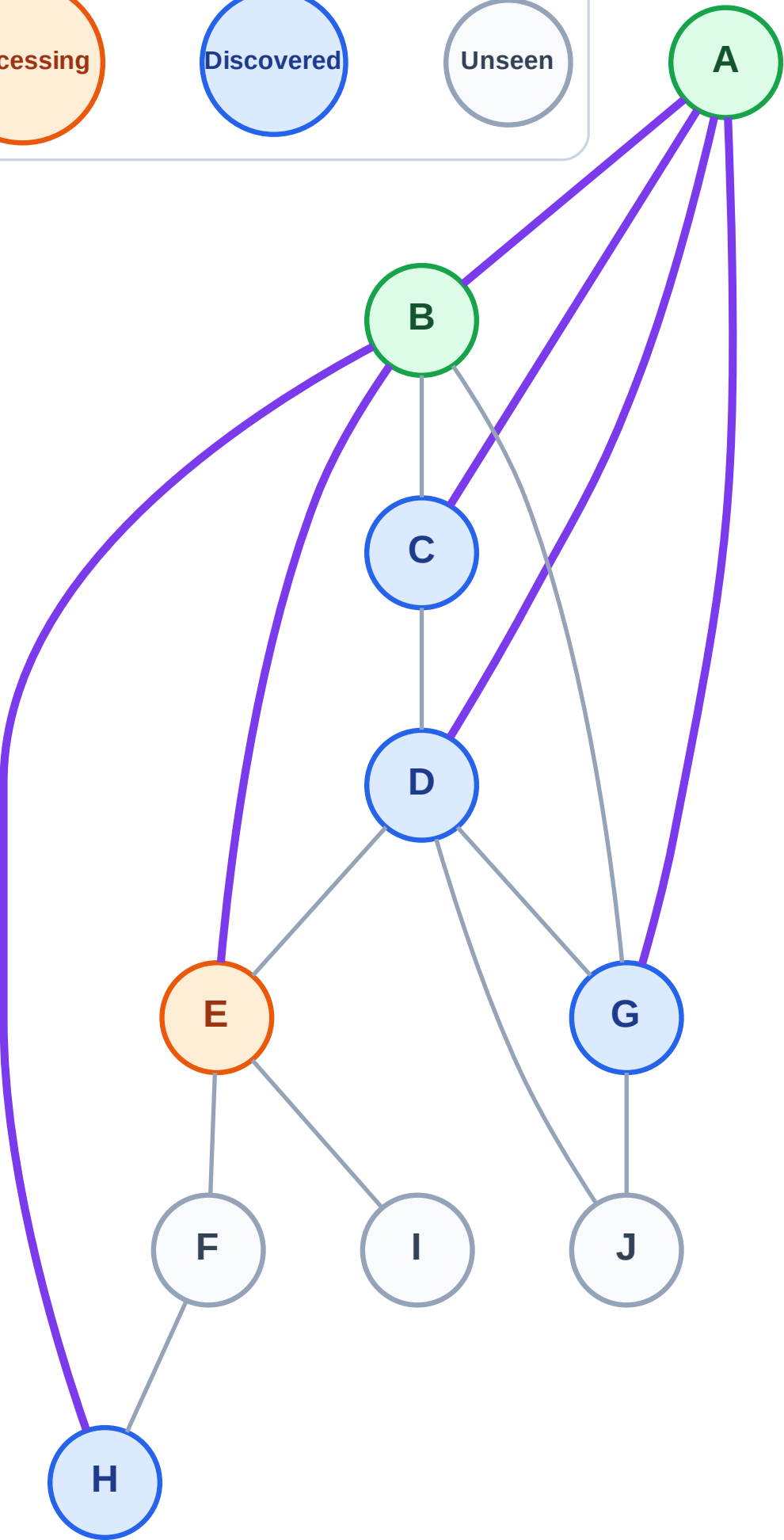
Processing

Discovered

Unseen

Stack (top at right): G | D | C | H

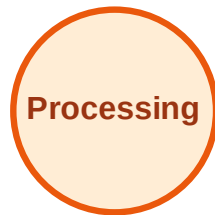
Visited: A, B



DFS Traversal

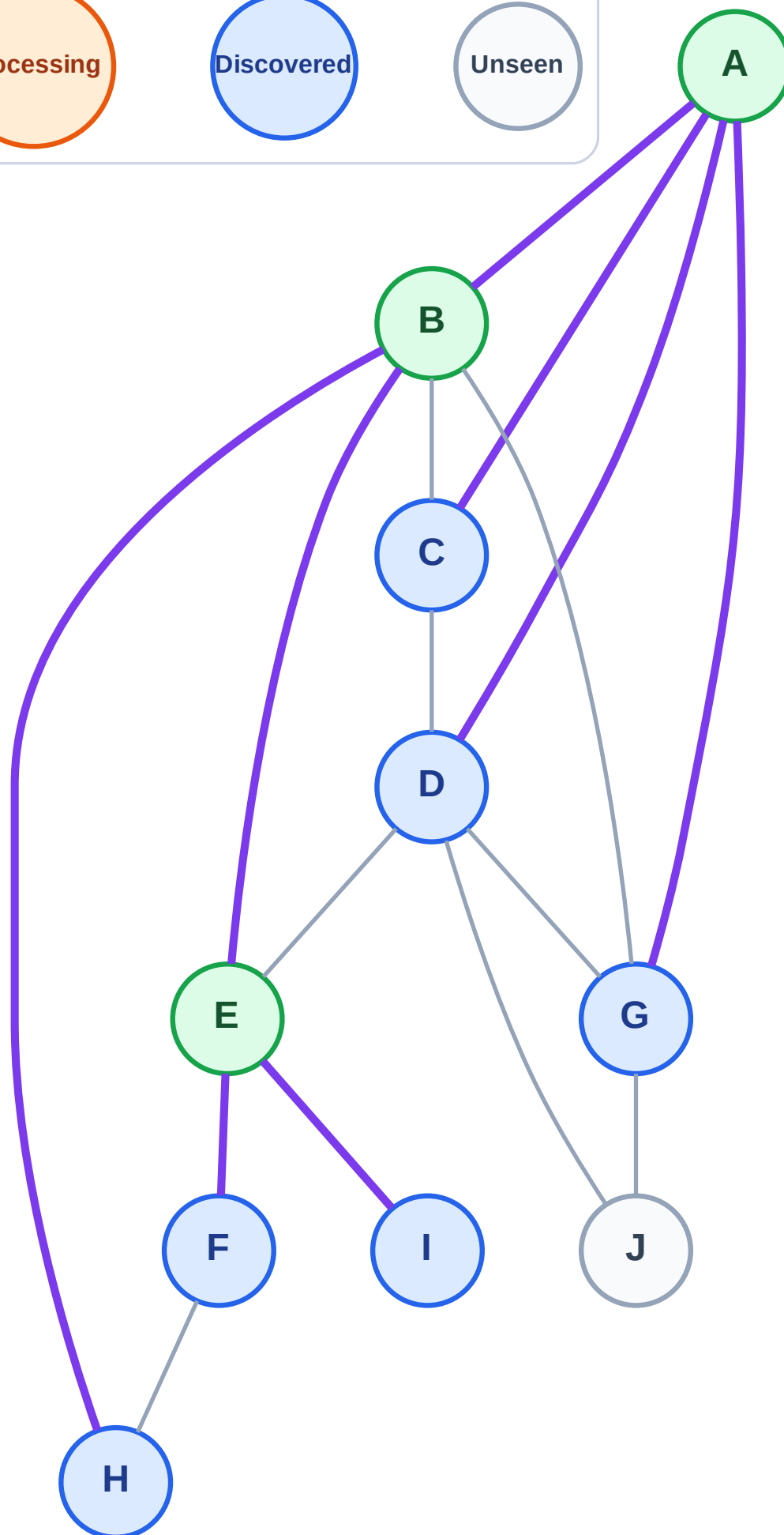
Step 7: Discover I, F from E

Legend



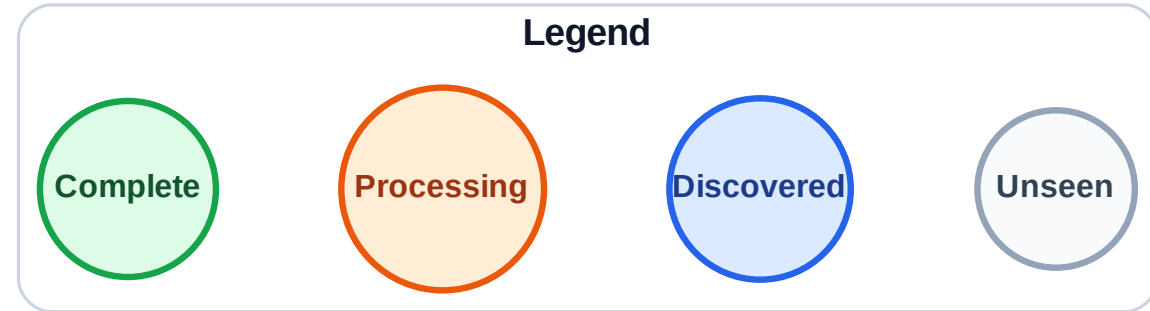
Stack (top at right): G | D | C | H | I | F

Visited: A, B, E



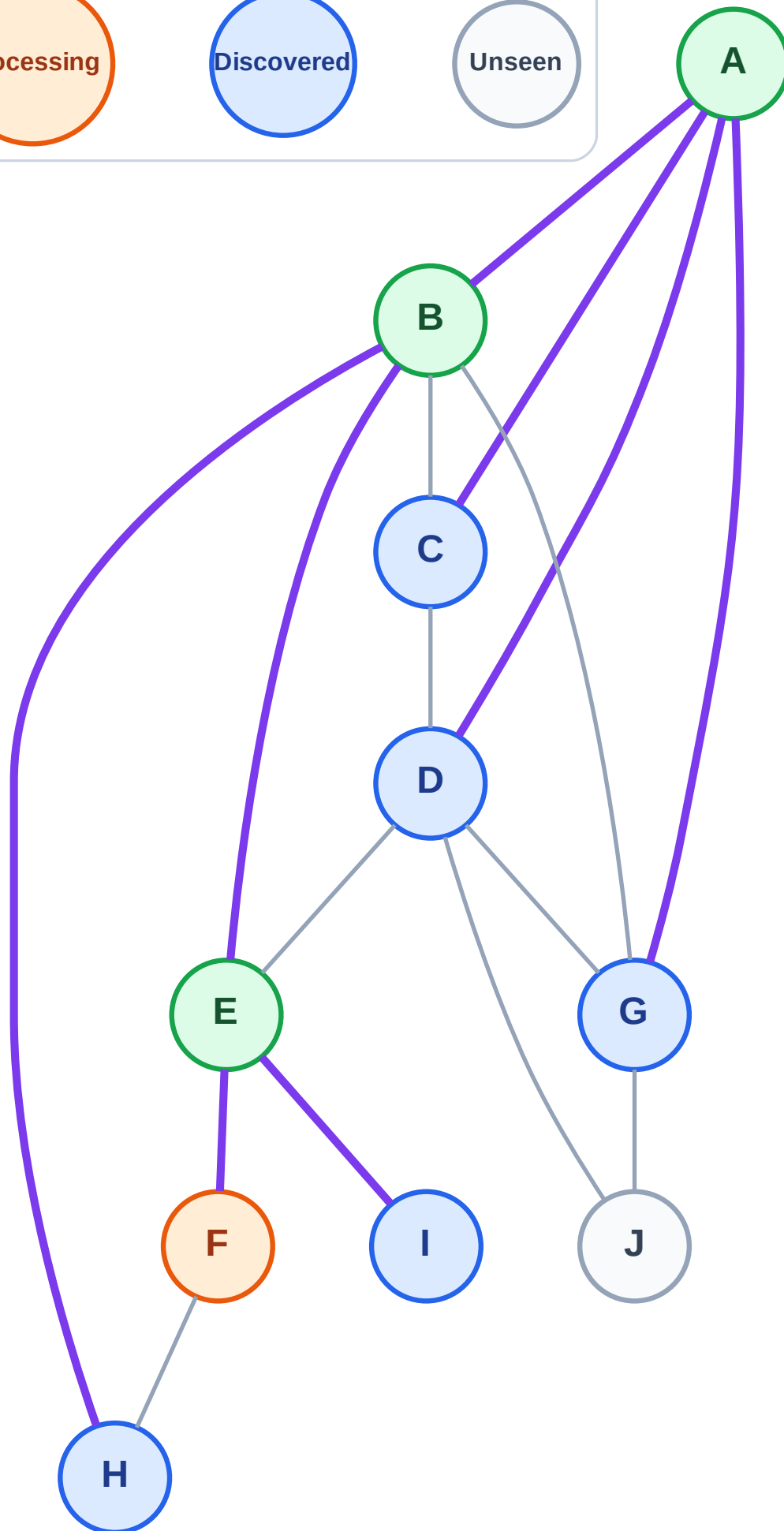
DFS Traversal

Step 8: Process node F



Stack (top at right): G | D | C | H | I

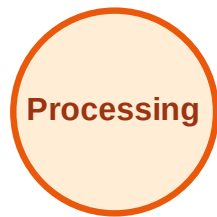
Visited: A, B, E



DFS Traversal

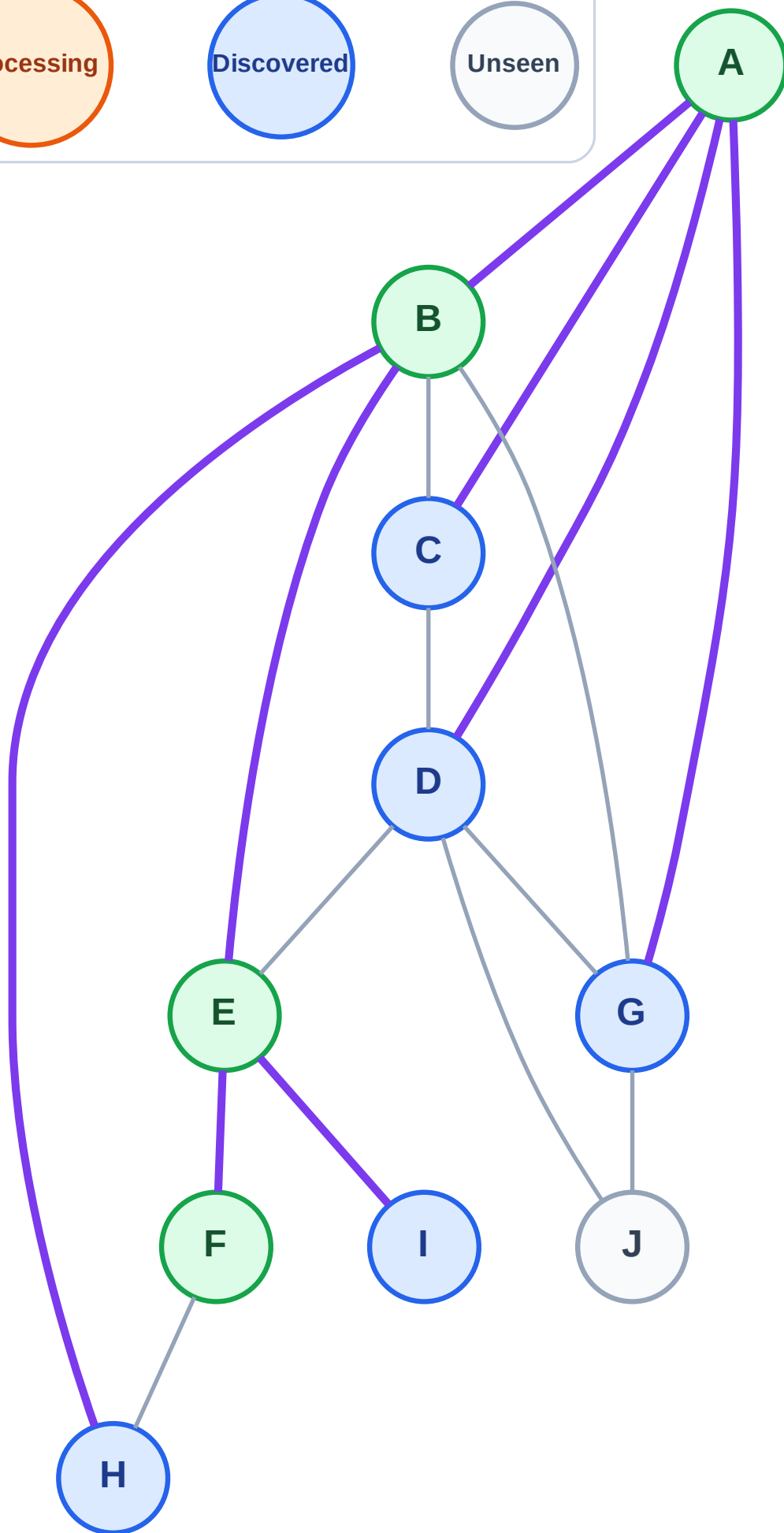
Step 9: No new neighbors from F

Legend



Stack (top at right): G | D | C | H | I

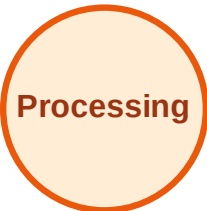
Visited: A, B, E, F



DFS Traversal

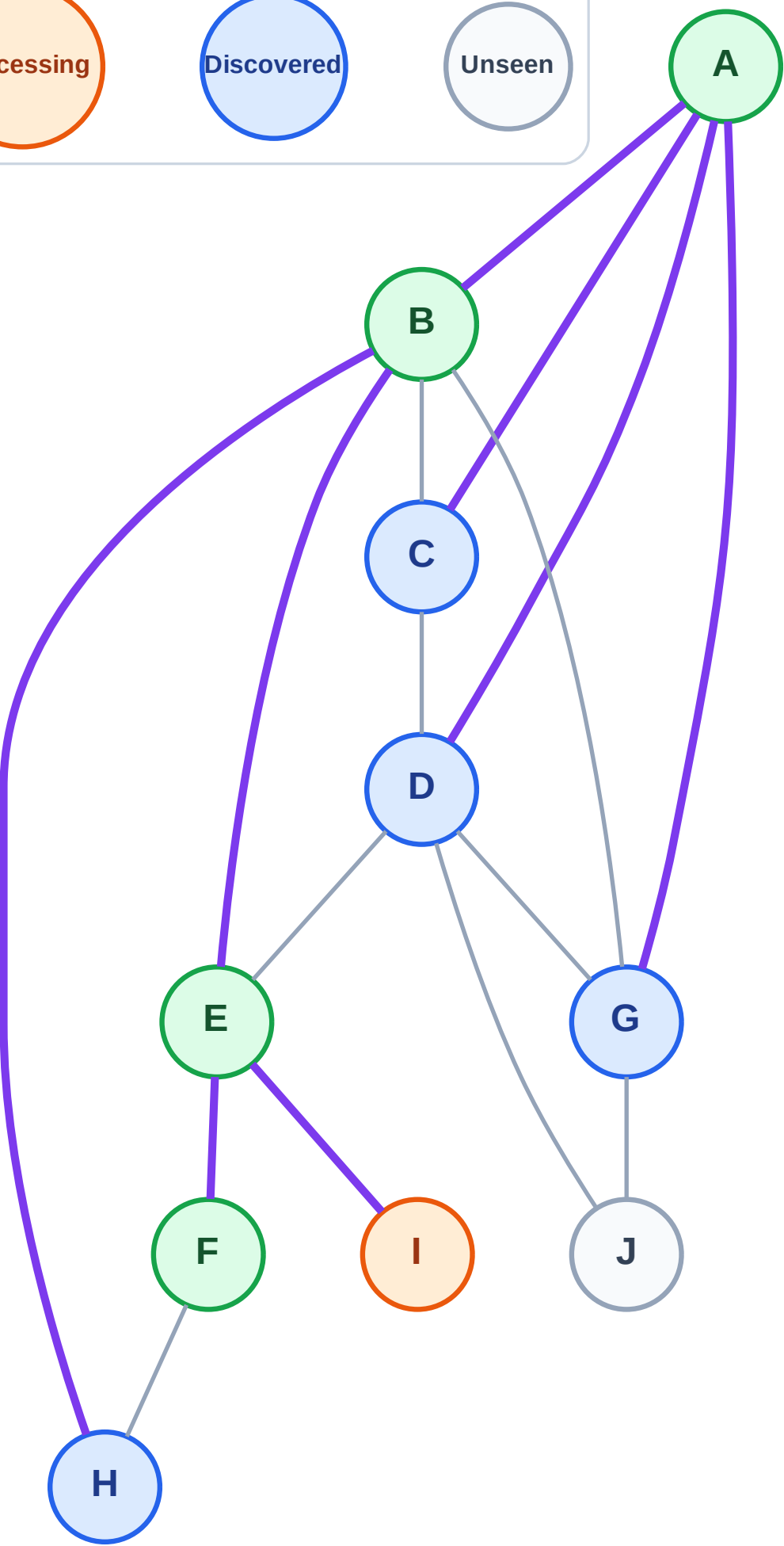
Step 10: Process node I

Legend



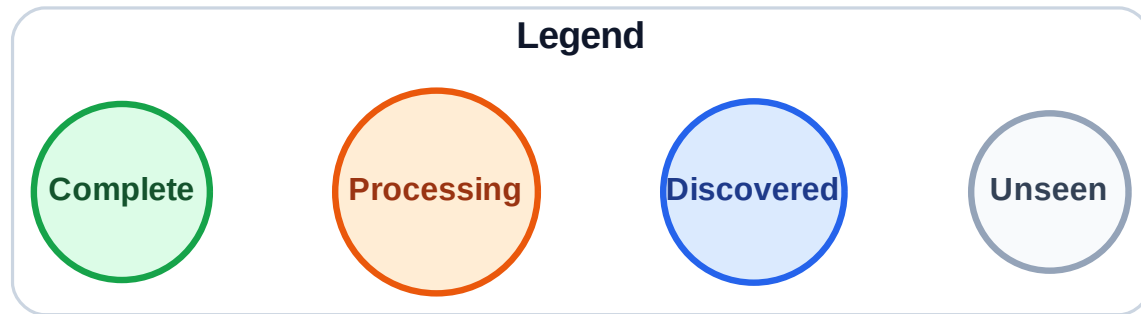
Stack (top at right): G | D | C | H

Visited: A, B, E, F



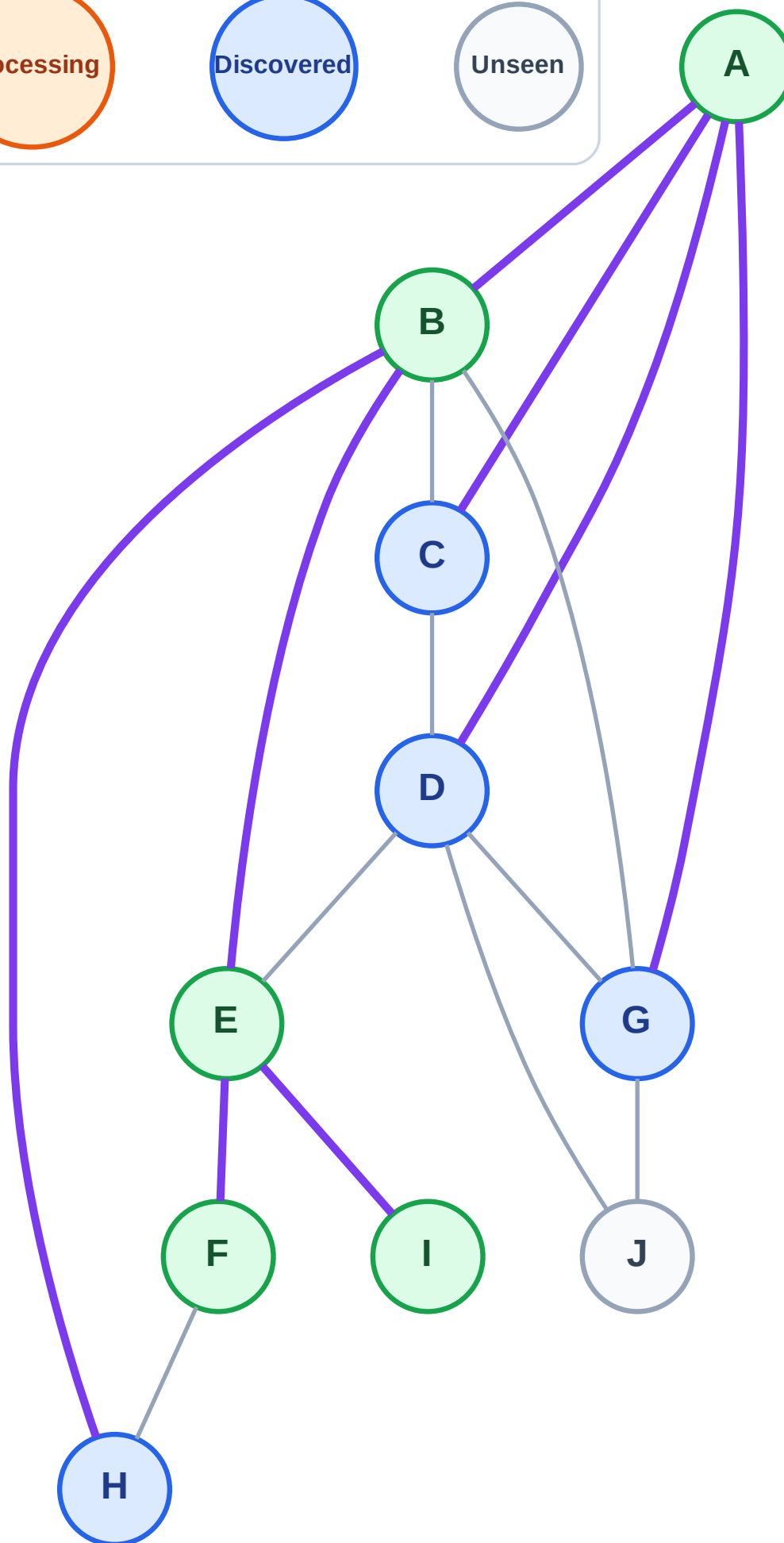
Step 11: No new neighbors from I

Step 11: No new neighbors from I



Stack (top at right): G | D | C | H

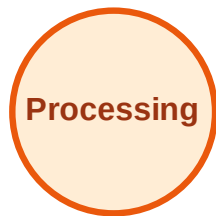
Visited: A, B, E, F, I



DFS Traversal

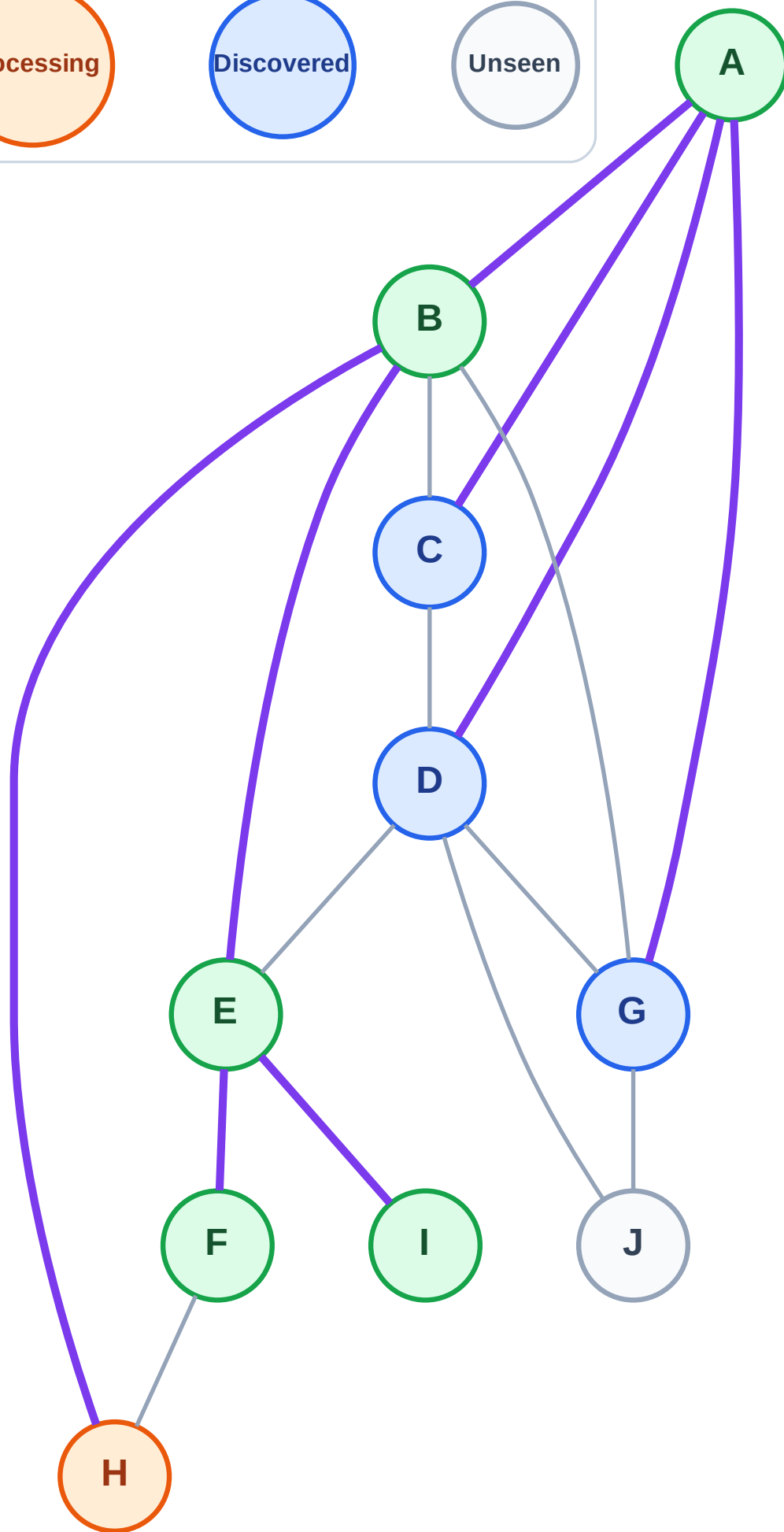
Step 12: Process node H

Legend



Stack (top at right): G | D | C

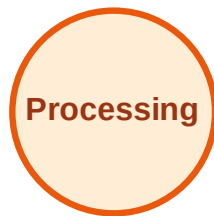
Visited: A, B, E, F, I



DFS Traversal

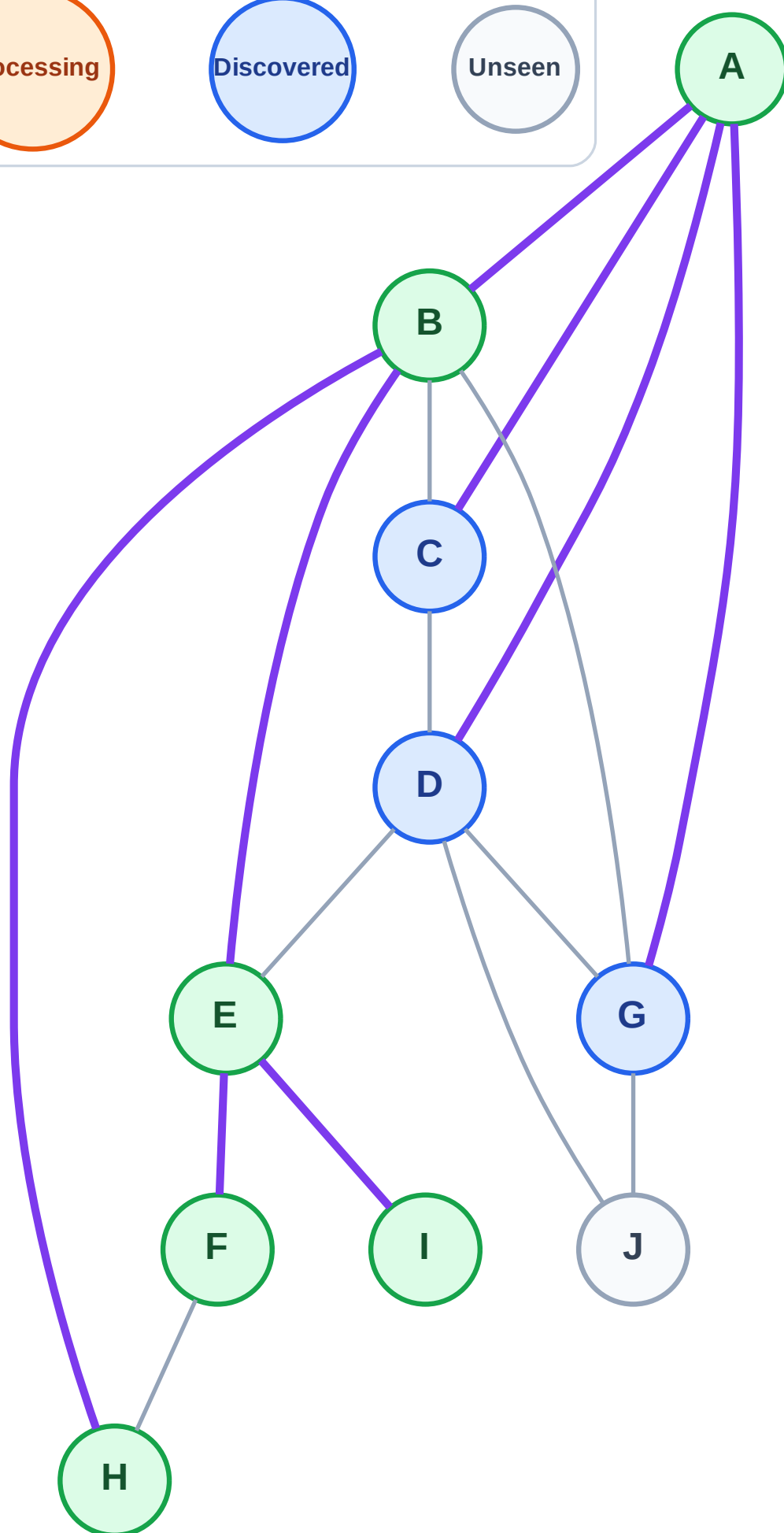
Step 13: No new neighbors from H

Legend



Stack (top at right): G | D | C

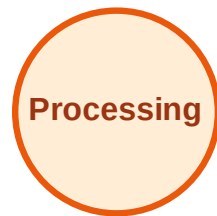
Visited: A, B, E, F, H, I



DFS Traversal

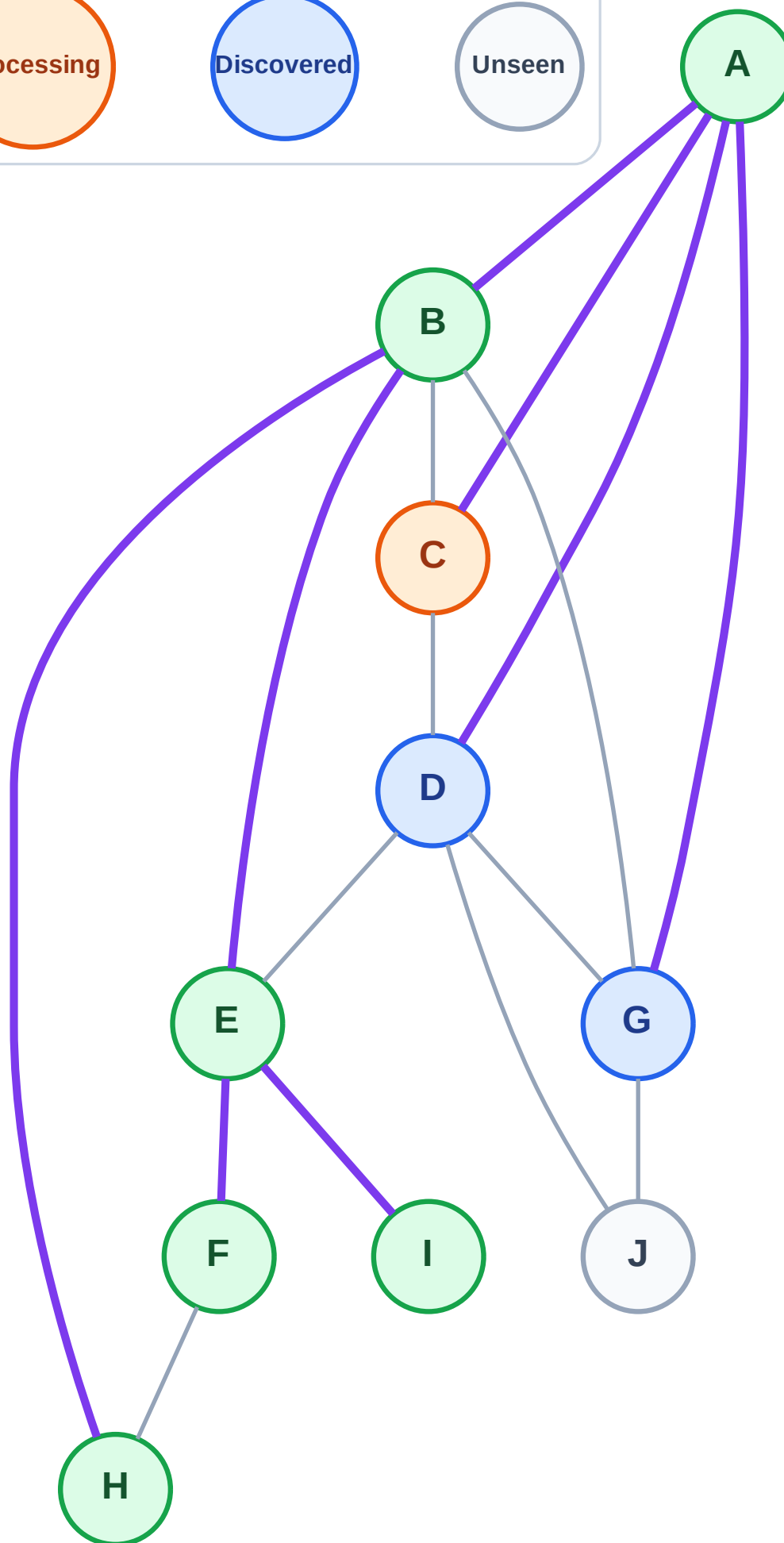
Step 14: Process node C

Legend



Stack (top at right): G | D

Visited: A, B, E, F, H, I



DFS Traversal

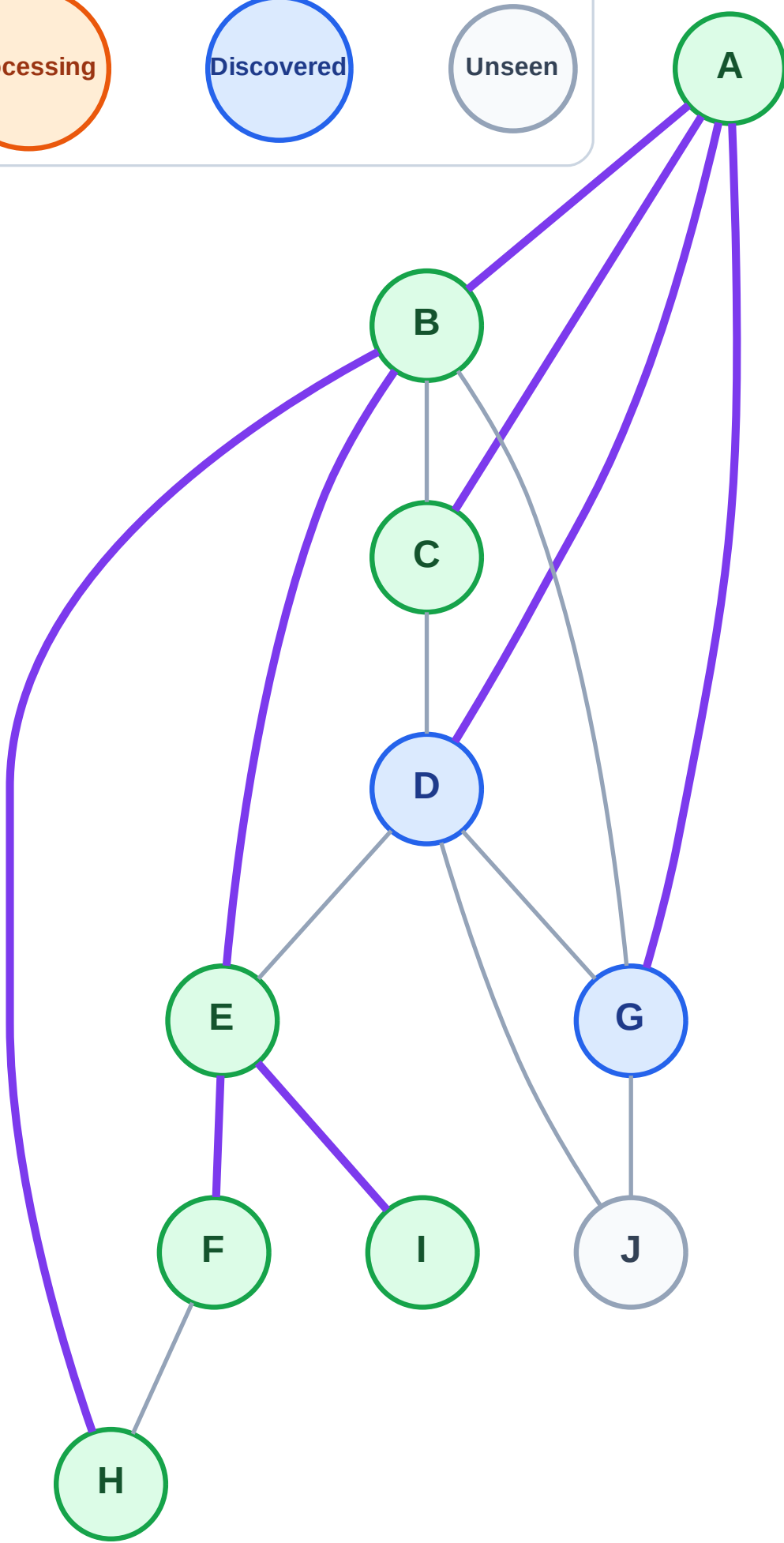
Step 15: No new neighbors from C

Legend



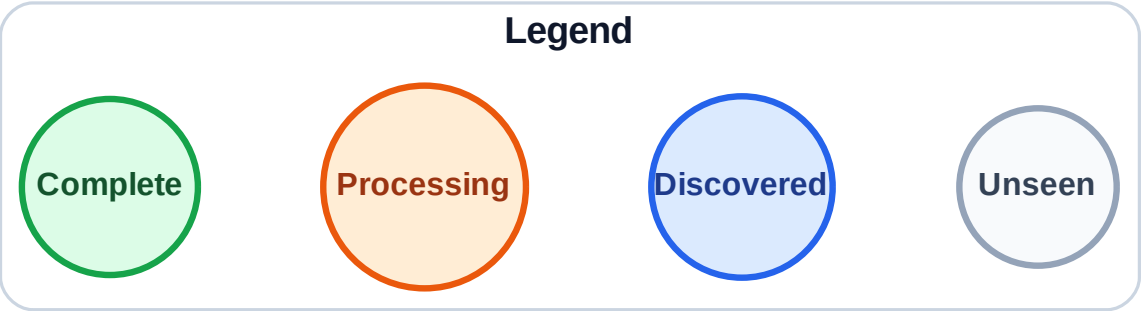
Stack (top at right): G | D

Visited: A, B, C, E, F, H, I

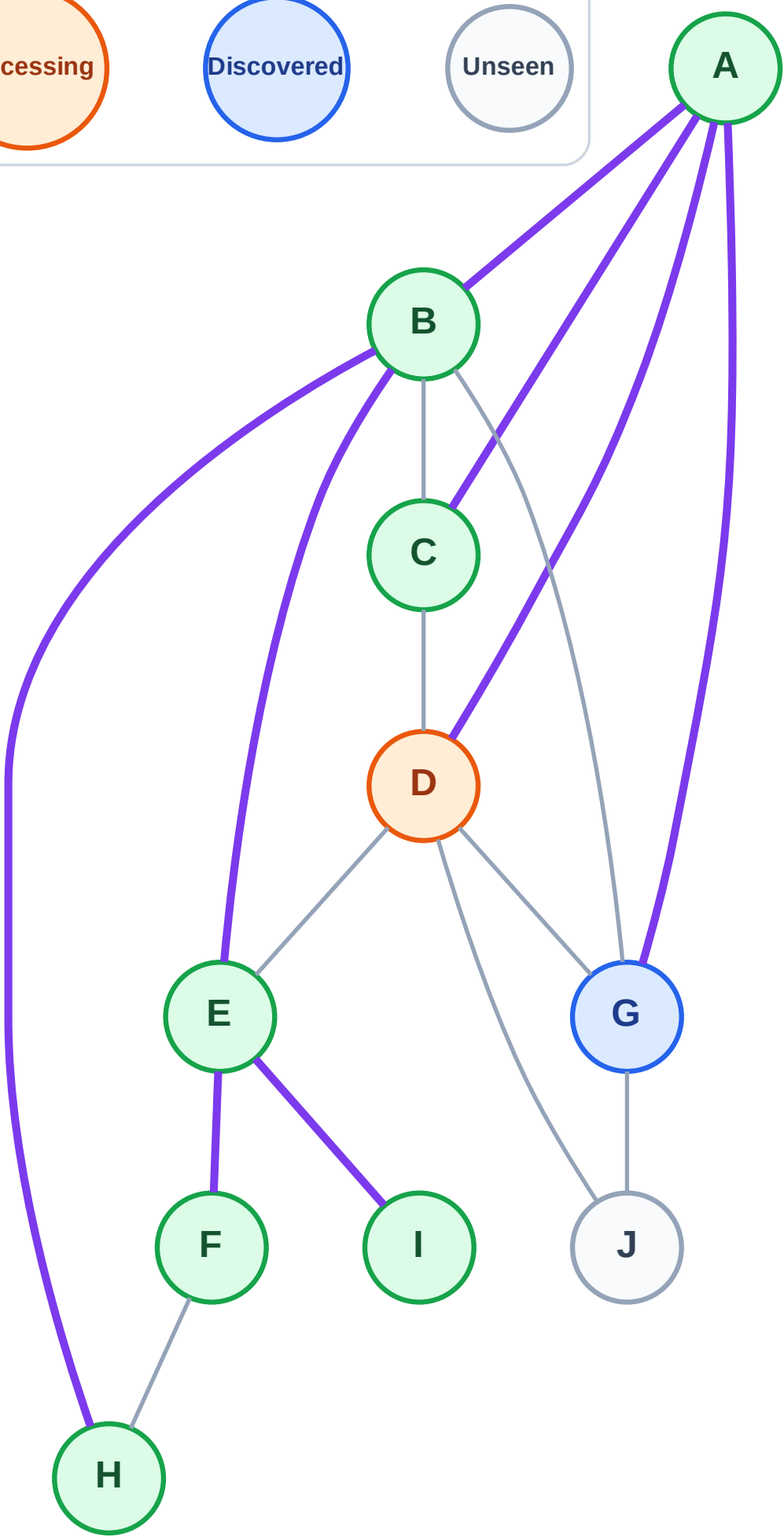


DFS Traversal

Step 16: Process node D



Stack (top at right): G
Visited: A, B, C, E, F, H, I



DFS Traversal

Step 17: Discover J from D

Legend

Complete

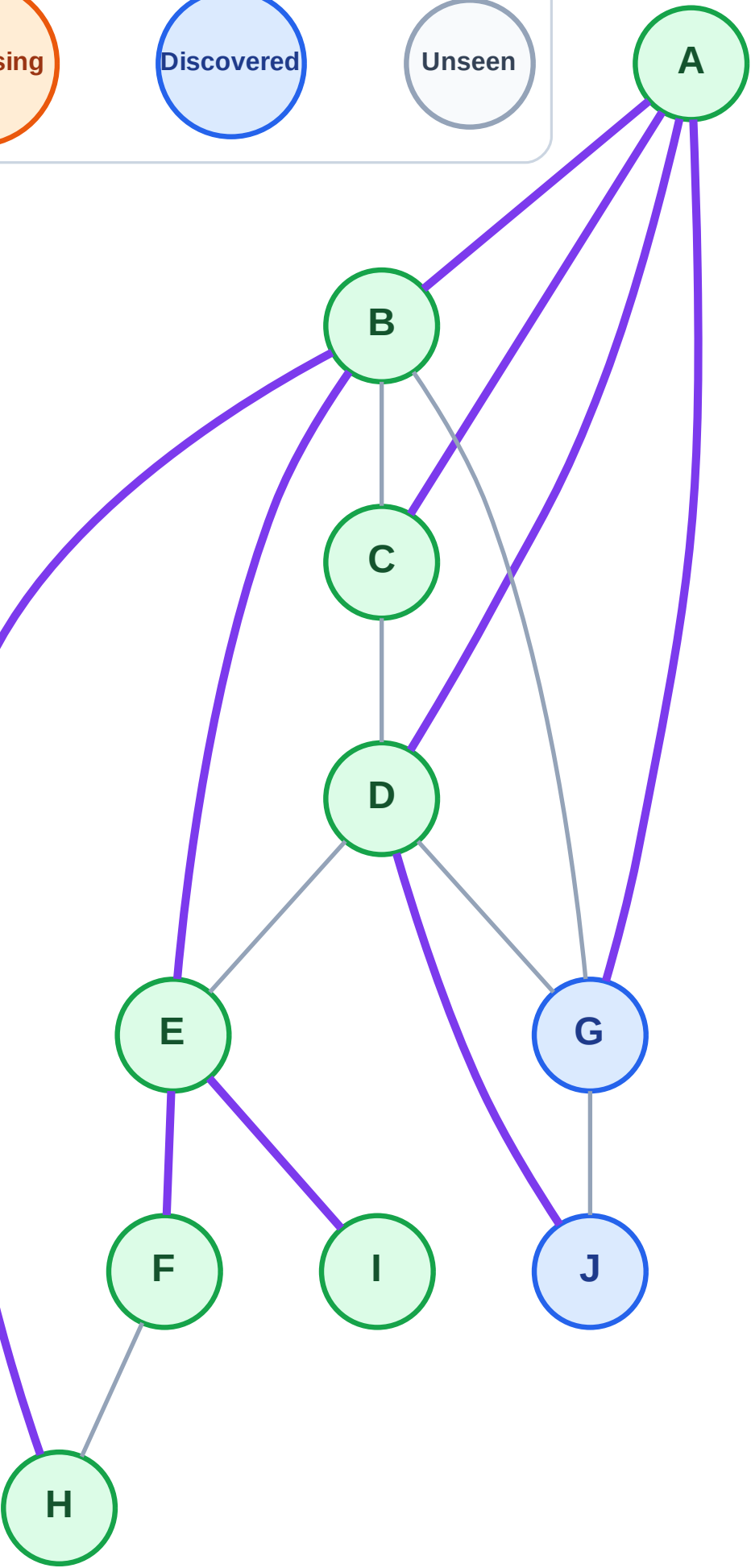
Processing

Discovered

Unseen

Stack (top at right): G | J

Visited: A, B, C, D, E, F, H, I



DFS Traversal

Step 18: Process node J

Legend

Complete

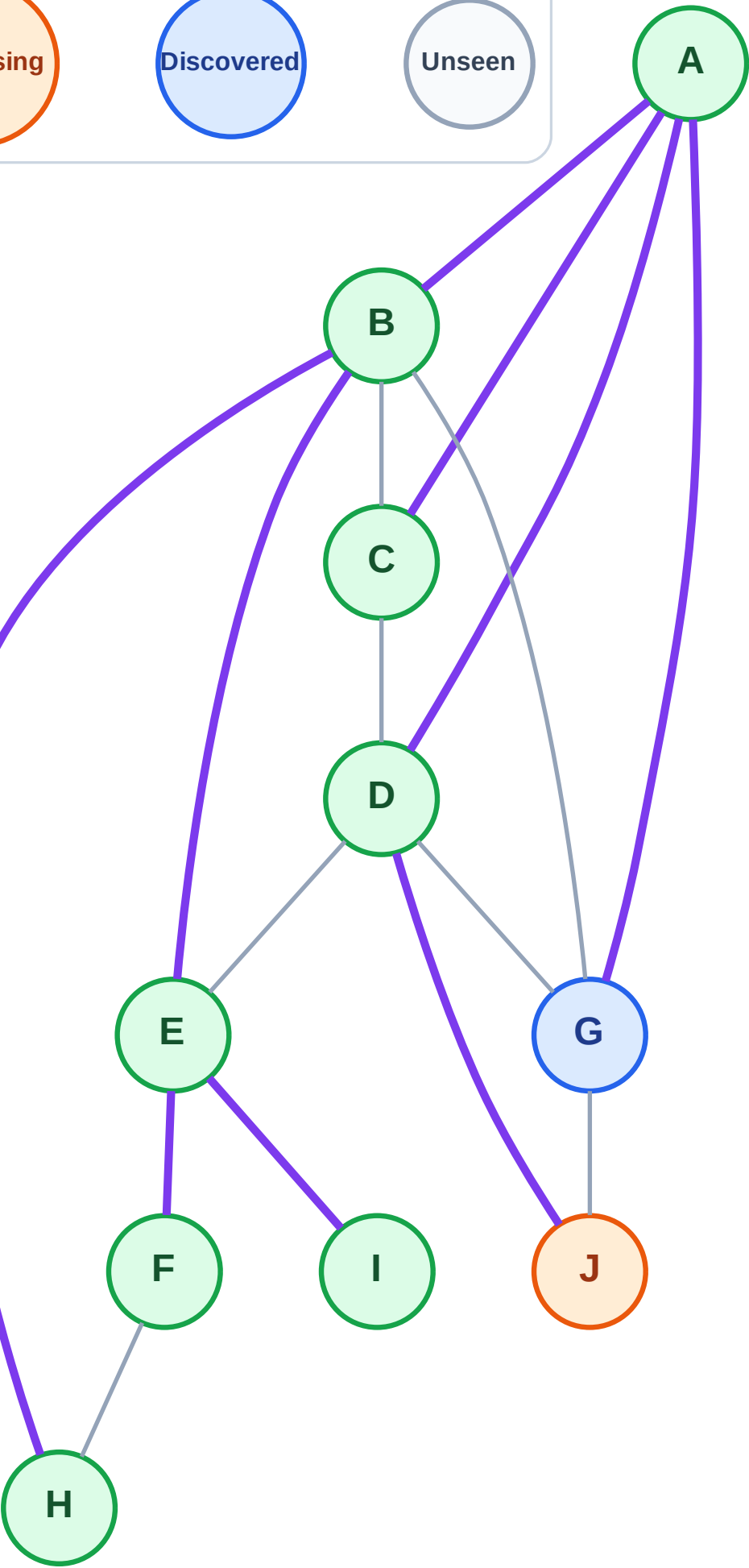
Processing

Discovered

Unseen

Stack (top at right): G

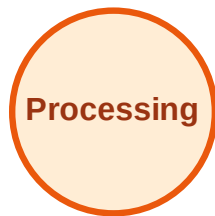
Visited: A, B, C, D, E, F, H, I



DFS Traversal

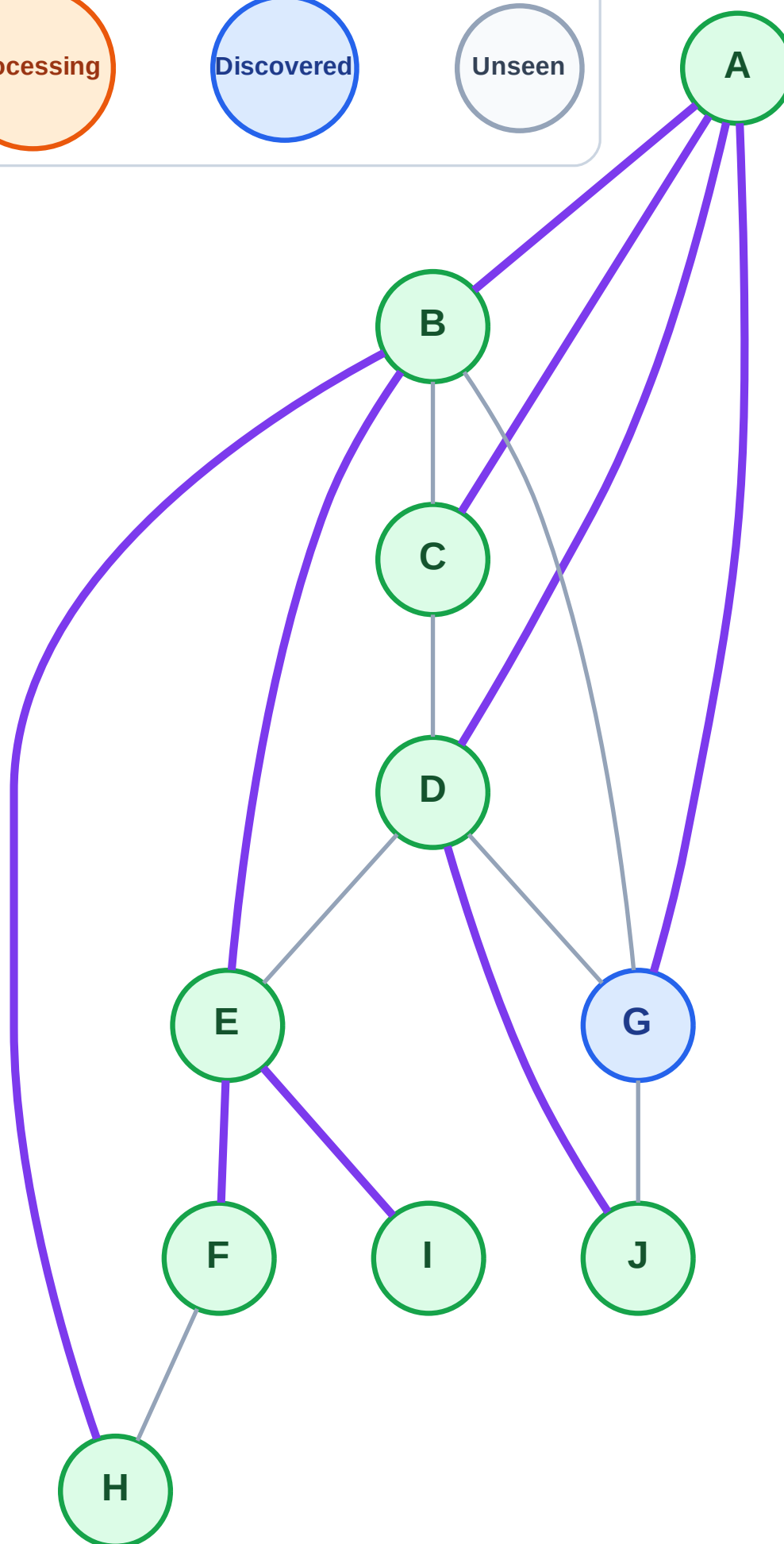
Step 19: No new neighbors from J

Legend



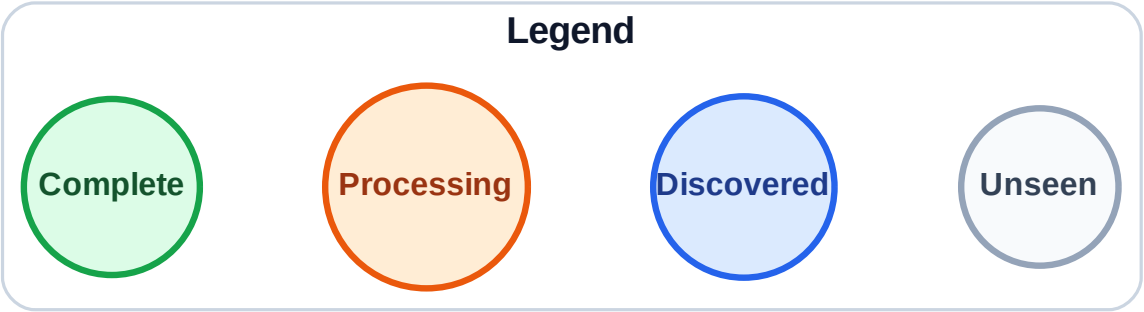
Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J

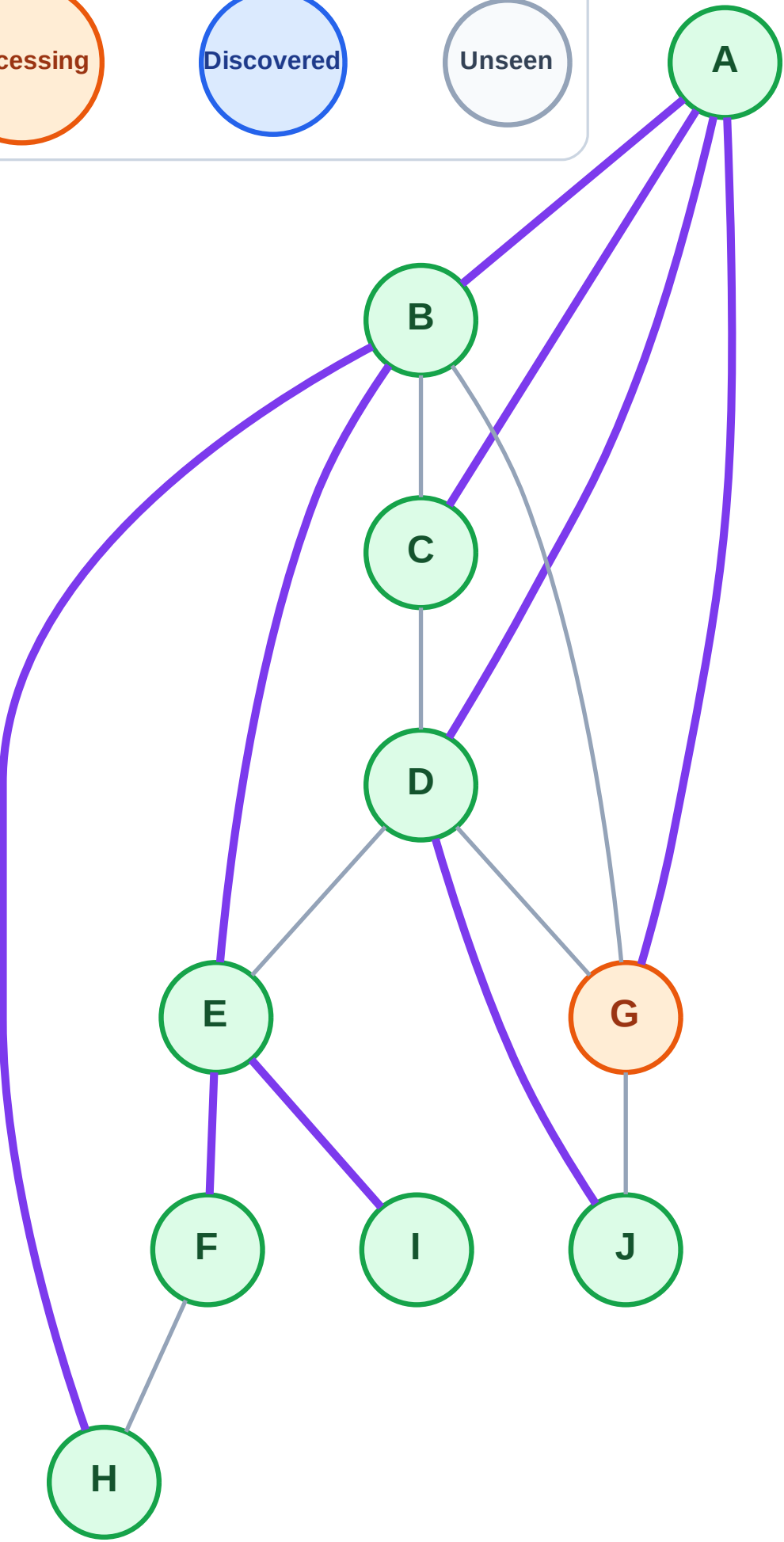


DFS Traversal

Step 20: Process node G



Stack (top at right): (empty)
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

